

CV - ROSALBA PERNA

Research Interests

High Energy Astrophysics

Gamma-ray bursts, neutron stars with special focus on magnetars, accretion disks, X-ray binaries, ultra-luminous X-ray sources, mergers of binary compact objects, gravitational waves and their electromagnetic counterparts, tidal disruption events.

Cosmology

Gravitational lensing, interstellar medium and dust in high-redshift galaxies, foregrounds for CMB and 21 cm measurements, growth of supermassive black holes, axions.

Exoplanets

Magnetohydrodynamics and turbulence in the atmospheres of hot Jupiters. Dynamical effects from stellar encounters on planetary systems.

N -body Dynamics

Scattering in the small- N regime with applications ranging from supermassive black hole dynamics in galactic centers, to production of hypervelocity stars, to binary compact object formation, to exoplanetary architectures in dense star clusters.

Education

Ph.D. in Physics

September 1994- June 1999

Harvard University, Cambridge, MA Thesis title: *Theoretical studies of Gamma-Ray Bursts*.

Piano Diploma

September 1993

Conservatorio di Musica, Potenza, Italy.

Laurea in Physics, 110/110 cum laude

November 1989–July 1993

Università di Salerno, Italy. Thesis title: *Theory of the Integer and Fractional Quantum Hall Effect*

Academic Appointments

Stony Brook University

September 2021 – August 2024

Associate Department Chair

Stony Brook University

September 2017 –

Professor of Physics and Astronomy

Stony Brook University

January 2014 – August 2017

Associate Professor of Physics and Astronomy

University of Colorado at Boulder

March 2011 – December 2013

Associate Professor of Astrophysics

University of Colorado at Boulder

September 2004 – March 2011

Assistant Professor of Astrophysics

Princeton University

September 2003 – August 2004

Lyman Spitzer Fellow

Harvard Society of Fellows

July 1999 – June 2003

Junior Fellow

Visiting Scientist Positions

Carnegie Observatories (01/2017); California Institute of Technology (04/2002–06/2002); Osservatorio Astronomico di Roma (01/2002–03/2002); Istituto di Astrofisica Spaziale, Italy (10/2001–12/2001); CCA, Flatiron Institute (2018–)

Awards and Honors

Aspen Institute Italia Award	2022
For significant US-Italian collaborative research	
OCIW Distinguished Scientific Visitor	2017
Carnegie Observatories	
Fellowship in the American Physical Society	2014
American Physical Society	
SIGRAV Prize	2010
Italian Society of General Relativity and Gravitational Physics	
Astronomy Magazine Feature	January 2010
Personal profile as an Astrophysicist featured in the Astronomy magazine	
Provost's Faculty Achievement Award	2009
University of Colorado at Boulder	
Kavli Frontiers of Science Fellow	2007, 2013
National Academy of Science and Kavli Foundation	
Lyman Spitzer Fellowship	September 2003 - August 2004
Department of Astrophysics, Princeton University	
Harvard Junior Fellowship	July 1999 - June 2003
Harvard Society of Fellows	
Award of Excellence (Top Graduate of 30 years)	2002
Faculty of Science, University of Salerno (Italy)	
Predoctoral Fellowship	August 1998 - May 1999
Smithsonian Astrophysical Observatory.	
Amelia Earhart Fellowship	September 1998 - May 1999
Zonta International Foundation	
AWIS Predoctoral Award	1998
Association for Women in Science Educational Foundation	
Goldhaber Prize (Top Theory Student)	1997
Physics Department, Harvard University	
Fellowship for Research Abroad	September 1996 – August 1998
Physics Department, Università di Salerno (Italy).	
Rudmose Fellowship	September 1995 – June 1996
Physics Department, Harvard University	
Top Student Award (National)	September 1984 – June 1998
ENAM (Italy)	

Teaching

Stony Brook University:

- ”WISE Honors Project/Thesis I”, Fall 2023
- ”Tutorial in Advanced Topics”, Fall 2021
- “Interstellar Medium”, graduate course, Fall 2016
- “Graduate Seminar II” (Nuclear, Particle, Astronomy), Spring 2014, Fall 2014, Fall 2017
- “Stars and Radiation”, for undergraduate science majors, Fall 2014, Fall 2016
- “Introduction to Astronomy: Stars and Galaxies”, for non science majors, Spring 2015, Fall 2015
- “Astronomy Laboratory”, for non science majors, Spring 2015, Fall 2015
- “Introduction to the Solar System”, for non science majors, Spring 2016, Fall 2019, Fall 2020, Fall 2024, Fall 2026
- “Modern Physics” (Recitation), Fall 2017
- “Introduction to Planetary Sciences”, for both science and non-science majors, Fall 2020, Fall 2021, Fall 2022, Fall 2024

University of Colorado:

- “Black Holes”, an undergraduate course for non-science majors (taught seven times)
- “Internal Processes in Gases, part II”, a graduate level course (taught twice)
- “Astrophysics I, Stellar and Interstellar”, for undergraduate science majors (taught once)
- “Stars and Galaxies”, for undergraduate non-science majors (taught twice)
- “Neutron Stars”, graduate seminar (taught once)
- “Radiative and Dynamical Processes”, a 4-credit core course for graduate students (taught three times)

Student and Postdoctoral supervision

Postdocs

- Matthew van Adelsberg, JILA, 09/2006 - 08/2008
- Bruno Giacomazzo, JILA, 10/2011 - 09/2013
- Takamitsu Tanaka, Stony Brook University, 09/2014 - 08/2016
- Simone Dall’Osso, Stony Brook University, 05/2016 - 02/2018
- Nathan Leigh, Stony Brook University, 01/2018 - 31/2018

Graduate Students (advised and co-advised)

- Joseph Flasher, University of Colorado, 2004-2006
- Kevin Heng, University of Colorado, 2005-2007

Enrico Bozzo, University of Rome and JILA, 2003-2006
Peter Robinson, University of Colorado, 2007-2009
Federico Bernardini, University of Rome and JILA, 2007-2010
Ting Yan, University of Colorado, 2009-2011
Daniele Viganó, University of Alicante and JILA, 2013
Sebastien Guillot, McGill University and JILA, 2013-2014
Taeho Ryu, Stony Brook University, 2014 - 2018
Yunan Xie, Stony Brook University (Master thesis), 2020
Yihan Wang, Stony Brook University, 2017 - 2022
Jason Alford, Columbia University, 2018 - 2021
Michael Ray, Stony Brook University (Master thesis), 2021, 2022
Claudia Soriano Guerrero, University of Barcelona, 2021 - 2025
Marguerite Epstein-Martin, Columbia University, 2021 - 2026
Chengcheng Xin, Columbia University, 2021 - 2023
Chaitanya Prasad, Stony Brook University, 2022 - 2023
Alex Laundry, Stony Brook University (Master research), 2023
Gaia Faj, Niels Bohr Institute, 2024
Hoyoung Kang, Stony Brook University (Master thesis), 2024 - 2025
Ian Johnson, Stony Brook University, 2024 -
Jake Hassan, Stony Brook University, 2024 -
Nathan Walker, Oregon State, 2024 -
Emily McPike, CUNY, 2025 - 2026

Undergraduate Students

Michelle Villeneuve, University of Colorado, 2012-2013
Shawn Hoose, Stony Brook University, 2014 - 2015
Samantha Scibelli, Stony Brook University, 2015 - 2016
Rina Lim, Stony Brook University, 2016-2017
Antonio Frigo, Stony Brook University, 2020-2021
Braeden Bierwiler, Stony Brook University, WISE program, Honors thesis, 2021
Anna Kovalskaya, Stony Brook University, WISE program, Honors thesis, 2023
Zhehuze Wang, Stony Brook University, 2026

Selected Invited Conference Talks, Lectures, Colloquia and Seminars

Invited Talk, “First National Meeting on GRBs”, Rome, Dec. 2001
Astronomy Colloquium, Caltech, April 2002

Invited Talk, “Gamma-Ray Bursts: The Brightest Explosions in the Universe”, Harvard University, May 2002

Invited Talk, 2nd Workshop “Gamma-Ray Bursts in the Afterglow Era”, Rome, Sept. 2002

Astrophysics Seminar, University of Santa Cruz, May 2002

Astrophysics Seminar, University of Santa Barbara, May 2002

Astrophysics Colloquium, University of Chicago, January 2003

Astrophysics Seminar, New York University, January 2003

Astronomy Colloquium, Columbia University, January 2003

Astrophysics Seminar, Harvard/CfA, January 2003

Astrophysics Seminar, Rutgers University, February 2003

Astrophysics Colloquium, University of Colorado, February 2003

Physics Colloquium, University of Rochester, March 2003

Astrophysics Colloquium, Stanford University, April 2003

Theoretical Astrophysics Seminar, UC Berkeley, April 2003

Astrophysics Seminar, Lawrence Livermore National Labs, April 2003

Astrophysics Colloquium, Princeton University, December 2003

Astrophysics Seminar, La Scuola Normale, Pisa, Italy, December 2003

La Scuola Normale Superiore, Dec. 2003; lectures for PhD students on the topic of “Accretion in Astrophysics”.

Invited Talk, “GRB Physics before Swift”, State College, PA, April 2004

La Scuola Normale Superiore, Dec. 2003; lectures for PhD students on the topic of “Accretion in Astrophysics”.

Astronomy Colloquium, Cornell University, February 2005

Astrophysics Colloquium, American Museum of Natural History, February 2005

Astrophysics Colloquium, Northwestern University, March 2005

Astrophysics Colloquium, University of Chicago, April 2005

Astrophysics Seminar, Rutgers University, April 2005

Invited Talk, , “Pulsar Wind Nebulae”, Harvard-Radcliffe, May 2005

Astronomy Colloquium, Harvard University, March 2006

Astrophysics Colloquium, Nanjing University, China, July 2006

Invited Talk in the session on “Gamma-Ray Bursts”, 36th COSPAR Scientific Assembly, Beijing, China, July 2006

Astrophysics Colloquium, McGill University, Montreal, Canada, October 2006

Theoretical Astrophysics Seminar, UC Berkeley, April 2007

Astronomy Colloquium, University of Nevada at Las Vegas, May 2007

Astrophysics Colloquium, University of Colorado, September 2008

Astronomy Colloquium, Columbia University, March 2009

Astronomy Seminar, Yale University, May 2009

Astrophysics Colloquium, American Museum of Natural History, May 2009

Astrophysics Seminar, New York University, October 2009

Physics Colloquium, University of Southern California, October 2009

ITC Colloquium, Harvard/ITC, October 2009

Astronomy Colloquium, Ohio State University, November 2009

Invited Talk, “From Planets to Galaxies”, Budapest, Hungary, June 2010

Invited Prize Talk at “XIX SIGRAV conference on General Relativity and Gravitational Physics”, Pisa, Italy, September 2010

Physics Colloquium, University of Missouri at Columbia, November 2010
 Invited Review Talk, “The Prompt Activity of Gamma-Ray Bursts”, Raleigh, NC, March 2011
 All JILA Colloquium, JILA, March 2011
 Astronomy Colloquium, University of Maryland, April 2011
 Astrophysics Seminar, Max Planck Institute for Gravitational Physics, Potsdam, Germany, June 2011
 Astrophysics Seminar, Institute of Space Sciences, Barcelona, Spain, September 2011
 Physics Colloquium, West Virginia University, October 2011
 High Energy Seminar, Max Planck Institute, Garching, Germany, November 2011
 Astronomy Colloquium, University of British Columbia, Vancouver, Canada, November 2011
 Astronomy Colloquium, ETH, Zurich, Switzerland, November 2011
 Invited Talk, “Planet-Z: atmospheres and interiors of exoplanets”, Zurich, Switzerland, November 2011
 Astrophysics Seminar, Rutgers University, December 2011
 Astronomy Seminar, Stony Brook University, February 2012
 ITC lunch talk, Harvard/ITC, March 2012
 Astrophysics Seminar, Center for Relativistic Astrophysics, Georgia Tech, March 2012
 Physics Colloquium, University of Barcelona, March 2012
 Astrophysics Joint Seminar, McGill University, Montreal, Canada, April 2012
 Astrophysics Seminar, North Carolina State University, Raleigh, April 2012
 Astrophysics Seminar, New York University, April 2012
 Astrophysics Seminar, Osservatorio Astronomico di Roma, Italy, June 2012
 Invited Talk in the session on “Magnetars”, 39th COSPAR Scientific Assembly, Mysore, India, July 2012
 Astrophysics Colloquium, Institut Astrophysique de Paris, September 2012
 Physics Colloquium, University of Texas at San Antonio, October 2012
 Cosmology Seminar, Carnegie Mellon University, December 2012
 Physics Colloquium, Grinnell College, December 2012
 Astrophysics Colloquium, Anton Pannekoek Institute, Amsterdam, Netherlands, December 2012
 Physics Colloquium, University of Massachusetts at Amherst, January 2013
 Astrophysics Seminar, La Scuola Normale, Pisa, Italy, February 2013
 Astrophysics Seminar, Stony Brook University, February 2013
 Astronomy Colloquium, Cornell University, April 2013
 Invited Talk, STARS2013 - SMFNS2013 meetings, Cuba, May 2013
 Astrophysics Colloquium, SISSA, Trieste, Italy, May 2013
 Invited Introductory Speaker for “Cosmic Explosions”, Kavli Frontiers of Science Symposium, National Academy of Sciences, Irvine, June 2013
 Invited Talk on the topic of ‘Magnetars’, KIPAC@10, “Big Questions in Particle Astrophysics and Cosmology”, Stanford, September 2013
 Invited Talk, “Explosive Transients: Lighthouses of the Universe”, Santorini, September 2013
 Astrophysics Colloquium, Northwestern University, October 2013
 Exoplanet Seminar, Kavli Institute, Cambridge, UK, November 2013
 Astrophysics Colloquium, Institute of Astronomy, Cambridge, UK, November 2013
 Invited Review Talk on ‘X-ray flares as clues of the central engine’, “GRB-Magnetar Thinkshop”, Bormio, Italy, January 2014
 Physics/Mathematics Joint Colloquium, New York University Abu Dhabi, January 2014

High Energy Seminar, University of Michigan, Ann Arbor, March 2014
 Physics Colloquium, Wayne State University, March 2014
 Invited Talk in the session on “The Transient Gravitational Wave Sky”, APS April Meeting, Savannah, April 2014
 Astronomy & Astrophysics Colloquium, University of Rochester, April 2014
 Physics Colloquium, University of Rochester, April 2014
 Astronomy Tea Talk, Caltech, May 2014
 Joint CSH-WP Colloquium, University of Bern, Switzerland, May 2014
 Invited Review Talk on ‘Progenitors of short Gamma-Ray Bursts’, “Gamma-Ray Burst in the Multi-Messenger Era”, Paris, June 2014
 Invited Talk in the session on “Magnetars”, 40th COSPAR Scientific Assembly, Moscow, Russia, August 2014
 Invited Review Talk on ‘Gamma-Ray Bursts and Magnetars’, “SWIFT 10 Years of Discovery”, Rome, December 2014
 ESO/MPA/MPE/USM Joint Astronomy Colloquium, Garching, Germany, January 2015
 Astronomy Colloquium, Penn State University, February 2015
 Physics Colloquium, Brookhaven National Laboratory, March 2015
 Institute of Advanced Studies - Princeton University Joint Astrophysics Colloquium, April 2015
 Invited Plenary Talk at the NewCompStar Meeting, Budapest, Hungary, June 2015.
 Invited Review Talk on “Highly magnetized neutron stars: theory, observations and connection with gamma-ray bursts” at the Fourteenth Marcel Grossmann Meeting on General Relativity and Relativistic Astrophysics, Rome, Italy, July 2015
 Invited Talk in “The Zoo of Accreting Compact Objects”, Lorentz Center of Leiden, Netherlands, August 2015
 Astronomy Colloquium, University of Massachusetts at Amherst, October 2015
 Invited Talk on “Magnetar Formation” at the ISSI workshop on Relativistic Compact Objects, Bern, Switzerland, November 2015
 Astrophysics Colloquium, KIPAC/Stanford, December 2015
 Astronomy Colloquium, Yale, December 2015
 Physics Colloquium, Stony Brook University, April 2016
 Physics Colloquium, Oregon State University, November 2016
 Astrophysics Colloquium, Carnegie Observatories, January 2017
 BWIS (Brookhaven Women in Science) Colloquium, Brookhaven national Laboratory, February 2017
 Astrophysics Seminar, American Museum of National History, February 2017
 Physics Colloquium, New York University, April 2017
 Physics Colloquium, Texas Tech., April 2017
 Invited Talk on “GRBs from compact-binary mergers” at “New Frontiers in Gravitational-Wave Astrophysics”, Rome, Italy, June 2017
 Invited Talk at the workshop “A multi-messenger look at the origin of gamma-ray busts”, in “European Week of Astronomy and Space Science”, Prague, Czech Republic, June 2017
 Invited Discussion Starter Talk on “Heating a long-lived dead disk” at the Lorentz Workshop on “Potential Electromagnetic signatures of stellar mass binary black hole mergers”, Leiden, Netherlands, September (2017)
 Invited Plenary Talk at the MAS Meeting of the American Physical Society, NJIT, November 2017
 Invited Plenary Talk on “EM counterparts of compact object binary mergers” at the conference “Deciphering the violent Universe”, Playa del Carmen, Mexico, Dec. 2017

Astrophysics Seminar, University of Waterloo, CA, March 2018
 Theoretical Astrophysics Seminar, UC Berkeley, April 2018
 Harvard-CfA Colloquium, April 2018
 Invited Talk in conference on "Unsolved Problems in Astrophysics and Cosmology", Budapest, Hungary, July 2018
 Astrophysics Seminar, Osservatorio Astronomico di Roma, Italy, June 2018
 Astrophysics Seminar, CITA, Toronto, Canada, November 2018
 Astronomy Seminar, Michigan State University, February 2019
 Astronomy Colloquium, University of Michigan, February 2019
 Invited Talk, "Fifty-One Ergs" conference, Raleigh, NC, May 2019
 Invited Talk at the "Multi-Messenger Astrophysics in the Gravitational Wave Era", Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Kyoto, Japan, October 2019
 Physics Colloquium, Georgia Tech, November 2019
 Astrophysics Colloquium, Center for Computational Astrophysics, Flatiron Institute, November 2019
 Invited Talk at the workshop on "The Origins of Black Hole Mergers and Gravitational Waves", Lorentz center, Leiden, Netherlands, December 2019
 Physics Colloquium, City College of New York, November 2020 (virtual)
 Astrophysics Seminar, McGill Space Institute, Montreal, Canada, January 2021 (virtual)
 Astrophysics Seminar, University of Concepcion, Chile, May 2021 (virtual)
 Astrophysics Seminar, University of Milano Bicocca, Italy, June 2021 (virtual)
 High Energy Astrophysics seminar, UNAM, Mexico, June 2021 (virtual)
 Astrophysics seminar, Monash University, Melbourne, Australia, February 2022 (virtual)
 CIERA Astrophysics seminar, Northwestern university, May 2022
 Astrophysics Seminar, University of Milano Bicocca, Italy, June 2022
 Astrophysics Seminar, University of the Balearic Islands, Spain, December 2022
 Invited Talk at the conference "AGN Santa Fe: Where are the Objects in AGN Disks?", Santa Fe, March 2023
 Invited Talk at the workshop on "Tidal Disruption Events", Columbia University, March 2023
 Astronomy Colloquium, University of Wisconsin, November 2023
 Invited talk at the conference on "Fast-Evolving Extragalactic Transients", Bormio, Italy, February 2024
 Astrophysics Seminar, Purdue University, April 2024
 Physics Colloquium, Carnegie Mellon University, February 2025
 TAP Colloquium, University of Arizona, April 2025
 Invited Talk at the workshop "Rattle and Shine: Gravitational waves and more", Hamburg Observatory, September 2025
 Invited Talk at the conference "The Tenth Anniversary of the Discovery of Gravitational Waves", Palma, Spain, September 2025
 Invited Talk at the "Dark Sky Meeting", La Vid, Spain, August 2026
 Astrophysics seminar, New York University, September 2026

Grants

Reported only RP's amount when shared grant

NASA/Swift (theory): (PI) Constraining the Properties of the Gamma-Ray Burst Environment through time-variable absorption. Dates: 05/01/2005 - 04/30/2007. Award amount: \$40,000

NSF-AST: (PI) Gamma-Ray Bursts as Cosmological Probes. Dates: 07/01/2005 - 06/30/2008. Award amount \$277,707

NASA/XMM: (CoI) Unveiling the broad band properties of the fading transient AXP XTE J1810-197. Dates: 09/18/2006 - /09/17/2008. Award amount: \$39,044

NASA/Spitzer: (CoI) Gamma-Ray Burst Physics in the Spitzer/Swift Era. Dates: 06/13/2006 - 09/30/2008. Award amount: \$19,605

NASA/Hubble: (CoI) Searching for Light Echoes After the SGR 1806-20-Hyperflare. Dates: 04/01/2006 - 03/31/2009. Award amount: \$21,419

NASA/XMM: (CoI) Unveiling the quiescent emission properties of the fading transient AXP XTE J1810-197. Dates: 07/24/2008 - 07/23/2009. Award amount: \$33,500

NASA/XMM: (CoI) The Transient Bursting Anomalous X-Ray Pulsar in Westerlund 1. Dates: 08/17/2007 - 08/16/2009. Award amount: \$33,645

NASA/XMM: (CoI) Unveiling the Quiescent State Properties of the Transient AXP XTE J1810-197. Dates: 01/01/2008 - 12/31/2009. Award amount: \$33,626

NASA/XMM: (CoI) The post-awakening of the transient bursting anomalous X-ray pulsar in Westerlund 1. Dates: 07/24/08 - 07/23/2010. Award amount: \$16,919

NASA/XMM: (CoI) Prompt study of AXP outbursts with XMM. Dates: 01/01/2009 - 12/31/2010. Award amount: \$34,800

NASA/Chandra: (CoI) Prompt Study of Burst-Selected Outbursts from AXPs. Dates: 03/15/2009 - 02/08/2012. Award amount: \$20,727

NASA/XMM: (CoI) Unveiling the nature of the Period evolution of XTE J180-197. Dates: 08/13/2009 - 08/12/2011. Award amount: \$25,165

NASA/XMM: (CoI) XMM-Newton Observation of the new SGR 0501+4516 in its return to quiescence. Dates: 09/01/2009 - 08/31/2011. Award amount: \$26,376

NASA/XMM: (CoI) From Outburst to Quiescence: Monitoring the Transient X-ray Pulsar in Westerlund 1. Dates: 09/01/2009 - 08/31/2011. Award amount: \$25,553

NASA/Chandra: (CoI) Unveiling the nature of cyclic behavior in the period evolution of the Anomalous X-ray Pulsar XTE J1810-197. Dates: 11/02/2009 - 01/25/2012. Award amount: \$21,362

NASA/XMM: (CoI) Prompt study of magnetar outbursts with XMM-Newton. Dates: 05/21/2010 - 05/20/2012. Award amount: \$15,616

NASA/Chandra (DDT): (CoI) Unveiling the nature of cyclic behavior in the period evolution of the Anomalous X-ray Pulsar XTE J1810-197. Dates: 06/08/2010 - 06/07/2011. Award amount: \$6,000

NSF-AST: (PI) Studies of Gamma-Ray Bursts in the Swift Era. Dates: 08/15/2010 - 07/31/2015. Award amount: \$383,071

NASA/Chandra (archive): (PI) Constraining the magnetic field topology of magnetars through spectral-timing analysis of the quiescent emission of XTE J1810-197. Dates: 01/01/2011 - 02/28/2013. Award amount: \$45,000

NASA/Chandra (DDT): (CoI) Unveiling the nature of cyclic behavior in the period evolution of the Anomalous X-ray Pulsar XTE J1810-197. Dates: 10/26/2010 - 10/25/2012. Award amount: \$7,500

NASA/Chandra (DDT): (PI) Unveiling the nature of cyclic behavior in the period evolution of the Anomalous X-ray Pulsar XTE J1810-197. Dates: 02/10/2011 - 02/09/2013. Award amount: \$12,000

NASA/Chandra: (CoI) Prompt study of magnetar outbursts with Chandra. Dates: 09/06/2011 - 06/23/2015. Award amount: \$21,712

NASA/Chandra: (CoI) Unveiling the Nature of the Cyclic Behavior in the Period Evolution of the Anomalous X-ray Pulsar XTE J1810-197. Dates: 02/21/2012 - 02/20/2015. Award amount: \$18,910

NASA/Fermi (theory): (CoI) General Relativistic Magnetohydrodynamic Simulations of Neutron Star Binaries. Dates: 08/07/2012 - 08/18/2015. Award amount: \$80,000

NASA/Chandra: (CoI) Prompt Study of Magnetar Outbursts with Chandra. Dates: 04/30/2013 - 04/29/2016. Award amount: \$23,212

NASA/Chandra: (CoI) Prompt Study of Magnetar Outbursts with Chandra. Dates: 07/16/2014 - 07/15/2016. Award amount: \$26,595

NASA/Hubble: (CoI) GRB hosts and the search for missing star formation at high redshift. Dates: 01/01/2015 - 12/31/2017. Award amount: \$5,000

NASA/Chandra (archive): (PI) Spectral and timing analysis of the low magnetic field magnetar SWIFT J1822.3-1606: testing our understanding of the magnetar behaviour. Dates: 01/01/2015 - 02/29/2017. Award amount: \$45,000

NASA/Swift (theory): (PI) X-ray Flares in long and short GRBs: A numerical investigation of disk fragmentation in hyperaccreting disks. Dates: 08/01/2015 - 07/29/2017. Award amount: \$40,000

NASA/Swift (theory): (PI) Swift precursors to long GRBs: hidden signs of a newly-born, hyperaccreting magnetar? Dates: 08/01/2016 - 07/31/2017. Award amount: \$40,000

NSF-AST: (PI) Gamma-Ray Bursts and Magnetars: Astrophysical Connections and Probes of Fundamental Physics. Dates: 08/01/2016 - 07/31/2020. Award amount: \$405,000

NASA/Chandra: (CoI) Prompt Study of Magnetar Outbursts with Chandra and NuStar. Dates: 05/25/2017 - 05/24/2018. Award amount: \$25,100

NSF-AST: (PI) Compact Object Binaries in the Gravitational Wave Era. Dates: 08/01/2020-07/31/2025. Award amount: \$450,000

NASA-Fermi (theory): (PI) Connecting the properties of binary compact merger ejecta with the prompt emission of their jet-cocoon systems Dates: 01/01/2021- 12/31/2021. Award amount: \$60,000

NASA Astrophysics Decadal Survey Precursor Science: (Co-I) Probing The High-z Universe with a Large X-Ray Observatory. Dates: 12/19/2024 - 12/18/2027. Award amount: \$283,582

NASA-Swift (theory): (PI) Calibrating the Stellar Population in the Disks of Active Galactic Nuclei via GRB Observations. Dates: 08/01/2025 - 07/31/2026 Award amount: \$35,000

Selected Approved Computational and Observing Proposals (as Co-I)

PRACE computer time grant: (PI: Giacomazzo) General Relativistic Simulations of Binary Neutron Star Mergers, 16 million core hours, 2015-2016

CINECA computer time grant: (PI: Giacomazzo) General Relativistic Magnetohydrodynamic Simulations of Neutron Stars on FERMI, 1 million core hours, 2014-2015

NSF XSEDE computer time grant: (PI: Giacomazzo) General Relativistic Magnetohydrodynamic Simulations of Merging Compact Objects, 4 million core hours, 2013-2014

NSF XSEDE computer time grant: (PI: Giacomazzo) General Relativistic Magnetohydrodynamic Simulations of Merging Compact Objects, 8 million core hours, 2012-2013

Chandra: Hot Accretion Flows in Quiescent Dwarf Novae

Hubble: Searching for light echoes after the SGR 1806-20- hyperflare

Spitzer: Gamma-Ray Bursts in the Spitzer/Swift Era

NOAO: Searching for Gamma-Ray Bursts in Nearby Galaxies

VLT: Probing Interstellar and Intergalactic matter through *UVES ToO* observations of GRB afterglows

Parkes: Searching for radio pulsations from the transient AXP in Westerlund 1

ALMA: Disentangling the nature of the ring-like structure around the magnetar SGR 1900+14

Professional Service - National and International

Referee for *The Astrophysical Journal*, *The Astrophysical Journal Letters*, *Astronomy & Astrophysics*, *Monthly Notices of the Royal Astronomical Society*, *Publications of the Astronomical Society of Japan*, *Publications of the Astronomical Society of Australia*, *International Journal of Modern Physics D*, *Astrophysics and Space Sciences*, *Physical Review Letters*, *Physical Review D*, *Physics of Fluids*, *Science*, *Nature*, *Nature Astronomy*, *Icarus*.

Hubble Cycle 12 (2003), Cycle 13 (2004), member of panel review

Swift Cycle 1 (2004), member of panel review

Columbia University (2004), external committee member for PhD thesis defense

Chandra Cycle 7 (2005), Cycle 9 (2007), Cycle 11 (2009), member of panel review, Cycle 17 (2015), Chair and member of panel review

National Science Foundation (2006, 2011, 2014, 2022, 2026), member of panel review

NASA-ATP (2006) and (2010), member of panel review

France-Israel joint astrophysics projects (2007), external reviewer

Niels Bohr Institute (2009), external committee member for PhD thesis defense

Fermi Cycle 2 (2009), member of panel review

NASA Postdoctoral Program (2010, 2012, 2013, 2014, 2023, 2024, 2026), reviewer

University of Stockholm (2011), member of international board for selection of new faculty member

Natural Sciences and Engineering Research Council of Canada (2012), reviewer of Major Resources Support Grants

Suzaku Cycle 7 (2012), member of panel review and Deputy Chair

National Science Center of Poland (2012), reviewer of grant proposals

Canada Research Chairs, member of the College of Reviewers for Senior Faculty Chair positions

University of Iceland (2013), external reviewer of postdoctoral grants

Einstein Postdoctoral Fellowships, member of the selection committee

University of Texas, San Antonio (2014-2018), external member of research committee panel for Physics PhD student

NASA Earth and Space Science Fellowships (2014), reviewer of proposals

La Scuola Normale di Pisa, reviewer of research proposals by their faculty

NSF EPSCoR Research Infrastructure Improvement (RII) Track-1 program, external reviewer of proposals

Netherlands Organization for Scientific Research (NWO), reviewer of Vidi grant application

Canadian Institute of Theoretical Astrophysics (CITA) (2015), member of the external Expert Review Committee in charge of evaluating the NSERC grant proposal of the Institute.

European Research Council (ERC) External reviewer of Consolidator (2016, 2017) and Starting Grant Proposals (2026)

Chilean National Science and Technology Commission (CONICYT) (2016), Reviewer of proposal for the FONDECYT grant competition.

Natural Sciences and Engineering Research Council of Canada (NSERC) (2016-2019), External reviewer of CITA yearly progress.

Advanced Theory and Simulations, Wiley (2017 –), member of the Editorial Board.

Universita' di Milano Bicocca (2018), reviewer of PhD thesis in Physics and Astronomy

NuSTAR Cycle 4 (2018), Chair and member of panel review

German Research Foundation, reviewer of grant proposal (2018)

Frontiers in Astronomy and Space Sciences, Associate Editor, 2018 - 2021

Kavli Institute for Theoretical Physics (KITP), Santa Barbara, Member of the Advisory Board, 2018 – 2022

NASA Multimessenger Astronomy, Member of the Science Advisory Group, 2018 –

Italian Ministry for Education, University and Research, referee for PRIN, to fund fundamental research projects.

Universita' La Sapienza, Rome, referee of PhD Thesis (2018)

NASA Hubble Fellowship program, Chair and panel member of the selection committee (2018).

CCA Summer School, “Multiscale Modeling of Astrophysical and Space Plasmas”, member of the organization and selection committee (2019)

International Conference on General Relativity and Gravitation (GR22), organizer and chair of the “Relativistic Astrophysics” session, Valencia, Spain, July 2019

F1000Prime, member of the section in High Energy Astrophysics, 2019 -

Universita' di Padova, Italy, referee of PhD Thesis (2019)

Blavatnik Regional Awards, member of the jury (2020)

National Academies of Sciences, Engineering, and Medicine (NASEM), Review Coordinator for the Astronomy and Astrophysics 2020 Decadal Survey (2020)

CCA, Flatiron Institute, co-leader of the Compact Object group (Fall 2020-Spring 2021)

CCA, Flatiron Institute, member of the selection committee for the Flatiron Research Fellows (2020)

Center for Astro Particle and Planetary Physics, NYU Abu Dhabi, member and Chair of the Center External Advisory Board (2021-2024)

Journal of High Energy Astrophysics (JHEAp), Associate Editor, 2021 – 2024

NASA Theoretical and Computational Astrophysics Networks (TCAN), Chair and member of the panel review

USA-Israel Binational Science Foundation, reviewer of proposal (2023)

NASA Exoplanets Research Program (XRP), member of the panel review

National Natural Science Foundation of China (NSFC) and the Israel Science Foundation (ISF), external reviewer of proposal

Raman Research Institute, external reviewer of faculty applicant (2023)

DCOMP/APS, member of the Nominating Committee (2023-2025)

German Research Foundation, reviewer of final project report for German-Russian collaborative grant (2023)

Universidad Nacional Autonoma de Mexico, member and president of the synod for PhD thesis evaluation and defense (2024)

Cottrell Scholar Award, reviewer of applications

Organization of Conferences and Schools

Member of the Scientific Organizing Committee for the parallel session on Gamma-Ray Bursts for the conference “New Views of the Universe”, Chicago, Dec. 2005.

Member of the Scientific Organizing Committee for the conference “Probing Stellar Populations Out to the Distant Universe”, Cefalu’ (Italy), Sept. 2008.

Member of the Scientific Organizing Committee for the conference “Gravitational Wave Bursts: Astrophysics, Data Analysis & Numerical Relativity”, Chichen-Itza (Mexico), Dec. 2009.

Member of the Scientific Organizing Committee for the School in Cargese on “Radiation Processes, Jets and Relativistic Objects”, Summer 2016

Member of the Scientific Organizing Committee for the session “X-ray Polarimetry: experiments and science prospects” at the 41st COSPAR Scientific Assembly, Istanbul, Turkey, July 2016

Member of the Scientific Organizing Committee for the session “The Magnetar Link in Neutron Stars, Gamma Ray Bursts and Supernovae” at the 41st COSPAR Scientific Assembly, Istanbul, Turkey, July 2016

Member of the Scientific Organizing Committee for the conference “FOE2017”, Corvallis, OR, June 2017

Member of the Scientific Organizing Committee for the mini-workshop on “Gravitational Waves and binary Neutron star Mergers”, Simons Center for Geometry and Physics, Stony Brook, November 2017

Member of the Scientific Organizing Committee for the mini-workshop on “The Plasma Physics of Neutron Stars Mergers”, Center for Computational Astrophysics, Flatiron Institute, October 2018

Member of the Scientific Organizing Committee of the session on “Neutron Star Astrophysics at the Crossroads: Magnetars and the Multimessenger Revolution” at IAU symposium 363, November 2021

Member of the Scientific Organizing Committee “Time Scales in Astrophysics”, Center for Astroparticle and Planetary Physics, New York University Abu Dhabi, January 2023

Member of the Scientific Organizing Committee of the Lorentz workshop “Challenges and future perspectives in gravitational- wave astronomy”, Leiden, October 2024

Member of the Scientific Organizing Committee “Cosmic Collisions”, Center for Astroparticle and Planetary Physics, New York University Abu Dhabi, January 2026

Departmental and University service

Stony Brook University:

Member of the interviewing committee for the incoming class of STEM scholars, 2024, 2025, 2026

Reviewer for the Workshop Seed Grant Program, 2025
 Member of the committee for the review of the Astro PhD requirements, 2024-2025
 Member of the Advisory committee of the Simons STEM Scholars Program, 2022
 Member of committee to select the Director of the Simons STEM Scholars Program, 2022
 Member of the search committee for a department Grant Administrator (2024)
 Coache University committee - Facilities and Infrastructure, 2023 -
 Graduate Committee, 2023 -
 Member of the standing committee for the P&A Master Program, 2022 -
 Chair and member of the Advisory Committee to the Chair, 2021 - 2024
 Member of the Department Webpage Committee, 2021 -
 Member of the Long Range Planning Committee, 2020-2021
 Member of promotion to full professor committee for colleagues in the department, 2018 –
 Member of PhD thesis committees, 2014 –
 Member of Master thesis committees, 2014 –
 Member of Colloquium committee, 2014, 2015, 2016, 2017, 2018
 Organizer of the astro seminars, Fall 2014, Spring 2015, Fall 2015, Spring 2016, 2023
 Member of the Qualifying Exam committee, 2014, 2015
 Member of Tenure committee for colleagues in the department, 2014 –
 Chair of committee for 4th year review of colleague in the department, 2015
 Member of the search committee for the Assistant Graduate Program Director, 2015
 Chair of the search committee for a faculty position in experimental high energy physics, 2015-2016
 Member of the search committee for a faculty position in Theoretical/Computational Astrophysics, 2016-2017
 Member of the search committee for a faculty position in Astrophysics/Large Data, 2016-2017
 Member of University committee for Diversity Plan, 2016, 2017, 2018
 Graduate Student Mentor, 2017-
 Member of the Chair search committee, 2017-2018
 Member of the Award committee, 2018-2020, 2022-

University of Colorado:

“Diversity and graduate students concerns” committee, 2004, 2005, 2006
 “Public relations and Newsletter” committee, 2004, 2005, 2006, 2007
 Colloquium organizer (chair), 2005, 2006
 Undergraduate advisor, 2006, 2007, 2009, 2010, 2012, 2013
 Graduate admission committee, 2006, 2007, 2008
 Comprehensive examination I committee, 2009, 2011, 2012, 2013
 Curriculum and undergraduate majors committee, 2009, 2012, 2013
 Course scheduling committee, 2010, 2011
 Committee member for Comprehensive Examination II (both Astrophysics and Physics Dept.), 2004-2013
 Committee member for PhD thesis defenses (both Astrophysics and Physics Dept.), 2004-2013
 JILA Executive committee member, 2013

Consortia and Professional Memberships

Invited Member of IAXO (International AXion Observatory)
 Invited Member of XIPE (X-ray Imaging Polarimetry Explorer); selected for ESA’s next medium-class science mission. Leader of the working group “Quantum electrodynamical effects and X-ray polarimetry”.
 Invited Member of the Advisory Board of the Collaborative Research Network on Neutron Stars, consisting of three Universities (Frankfurt, Jena, Frankfurt) and the Max-Planck Institute for Radioastronomy in Bonn.
 Scialog Fellow: Time Domain Astrophysics, selected participant

Member of the American Physical Society

Outreach and Volunteering

Piano teacher for low-income students, Salerno, Italy, 1991-1992

Reader for blind music student (music and text), Cambridge, MA, 1998-2001

Italian-English translator for Emergency USA, 2009

Panel member for discussion with undergraduate students at U. of Colorado on “Careers in Academia” (2009) and on “Women in Physics and Astronomy” (2013)

Faculty host at the Fiske Observatory night for the general public, Boulder (2010, 2012)

Volunteer for the “Earth Explorers” program for underrepresented youth (7th and 8th graders), Longmont, CO (2012)

Speaker in the “Open Night” series at Stony Brook University, Fall 2014

Guest expert on “Exoplanets” at the BNL astro journal club, Fall 2014

Volunteer for Achilles Int., for athletes (runners) with disabilities, Brooklyn, NY, 2018

Faculty member participant at the event “Wonder Women in Science and Engineering”, to inspire undergraduate women to a career in STEM, Stony Brook, October 2019, November 2020

Public Speaker at the Vanderbilt Planetarium for the Astronomical Society of Long Island, January 2020

Society of Physics Students talk, for undergraduate students interested in Physics, November 2020

WISE 105: Women in Science and Engineering seminar course, speaker, March 2021

WISE 102: Women in Science and Engineering research presentation, March 2021

Guest speaker for Science and Cinema, Stony Brook, April 2025

Personal

Citizenship: Italian and American citizen.

Hobbies: dance (argentine tango, salsa, swing, modern, jazz, aerial [reached level 3 in pole]), piano playing, travel, photography, bicycle touring, shaolin kung fu (reached 3rd degree brown belt), running (raced in numerous 1/2 marathons and marathons - 3 times Boston qualifier and 3 times NYC marathon qualifier).

Spoken languages: Italian (native), English (fluent), Spanish (proficient), French (basic).

Publications

Articles in refereed Journals

- 243. McPike, E., **Perna, R.**, et al., *McFACTS. IV. Electromagnetic Counterparts to AGN-disk-embedded Binary Black Hole Mergers*, *Astrophys. J.*, 1007, 105 (2026)
- 242. Cantiello, M., Hassan, J. B., **Perna, R.**, Armitage, P. J., Begelman, M. C., Jiang, Y.-F., Ryu, T., Townsend, R. H. D. *Pulsational Instability of Quasi-stars: Interpreting the Variability of Little Red Dots*, *Astrophys. J. Lett.*, 1000, L4 (2026)
- 241. Johnson, I. P. A., Ryu, T., **Perna, R.** *Hydrodynamic simulations of tidal disruption encores*, *Phys Rev. D.*, 113, 043032 (2026)
- 240. Hassan, J., **Perna, R.**, Cantiello, M., Armitage, P., Begelman, M., Ryu, T. *The Growth of the Central Black Holes in Quasi-stars*, *Astrophys. J.*, 998, 65 (2026)
- 239. Walker, N., Lazzati, D., Parsotan, T., **Perna, R.** *The Photospheric Emission of a Short-Duration Gamma-Ray Burst Emerging from a Realistic Binary Neutron Star Merger*, *Astrophys. J.*, 994, 93 (2025)
- 238. Hassan, J., **Perna, R.**, Cantiello, M., Parsotan, T., Lazzati, D., Walker, N. *Spectral Evolution of Rotating Population III Stars*, *Astrophys. J.*, 992, 68 (2025)
- 237. Viganó D., Sengupta, S., Soriano-Guerrero, C., **Perna, R.**, Elias-López, A., Kumar, S., Akgun, T. *Inflated hot Jupiters: inferring average atmospheric velocity via Ohmic models coupled with internal dynamo evolution*, *Astron. & Astrophys.*, 701, A8 (2025)
- 236. Hui, L. et al. (includ. **Perna, R.**), *Kilonova constraints for the LIGO/Virgo/KAGRA neutron star merger candidate S250206dm: GW-MMADS observations*, *Astrophys. J. Lett.*, 990, 46 (2025)
- 235. Soriano-Guerrero, C., Viganó, D., **Perna, R.**, Elias-Lopez, A., Beltz, H. *Non-ideal MHD simulations of hot Jupiter atmospheres*, *MNRAS*, 540, 1827 (2025)
- 234. Kang, H. D., **Perna, R.**, Lazzati, Davide, Wang, Yi-H. *The Cosmological Population of Gamma-Ray Bursts from the Disks of Active Galactic Nuclei*, *Open J. of Astrophys.*, 8, 23 (2025)
- 233. Wang, Y., **Perna, R.**, Zhu, Z., Lin, D. N. C. *Evolution of Extremely Soft Binaries in Dense Star Clusters: On the Jupiter Mass Binary Objects*, *Astrophys. J.*, 982, 90 (2025)
- 232. **Perna, R.**, Gottlieb, O., Shukla, E., Radice, D. *Connecting GRBs from Binary Neutron Star Mergers to Nuclear Properties of Neutron Stars*, *Phys. Rev. D*, 111, 063015 (2025)
- 231. Chimes, A. A. et al. (includ. **Perna, R.**), *The infrared counterpart and proper motion of magnetar SGR 0501+4516*, *Astr. & Astrophys.*, 696, 127 (2025)
- 230. Epstein-Martin, M., Tagawa, H., Haiman, Z., **Perna, R.** *Time-dependent models of AGN disks with radiation from embedded stellar-mass black holes*, *MNRAS*, 537, 3396 (2025)

229. Fabj, G., Dittmann, A. J., Cantiello, M., **Perna, R.**, Samsing, J. *Mapping the Outcomes of Stellar Evolution in the Disks of Active Galactic Nuclei*, *Astrophys. J.*, 981, 16 (2025)
228. Guglielmi, L., Stratta, G., Dall’Osso, S., Singh, P., Brusa, M., **Perna, R.** *Incidence of afterglow plateaus in GRBs associated with binary neutron star mergers*, *Astr. & Astrophys.*, 692, 73 (2024)
227. Lazzati, D., **Perna, R.**, Ryu, T., Breivik, K. *Delayed Emission from Luminous Blue Optical Transients in Black Hole Binary Systems*, *Astrophys. J. Lett.*, 972, 17 (2024)
226. Ascenzi, S., Viganó, D., Dehman, C., Pons, J. A., Rea, N., **Perna, R.** *3D code for MAgneto-Thermal evolution in Isolated Neutron Stars, MATINS: thermal evolution and lightcurves*, *MNRAS*, 533, 201 (2024)
225. Wang, Y., **Perna, R.**, Zhu, Z. *Free-Floating binary planets from ejections during close stellar encounters*, *Nature Ast.*, 8, 756 (2024)
224. Prasad, C., Wang, Y., **Perna, R.**, Ford, K. E. S., McKernan, B. *Tidal Disruption Events from three-body scatterings and eccentricity pumping in the disks of Active Galactic Nuclei*, *MNRAS*, 531, 1409 (2024)
223. Piga, V., Burgay, M., Possenti A., Ridolfi A., Pilia, M., Rea, N., **Perna, R.**, Colpi, M., Israel, G. *Serendipitous Discovery of a 431 ms Pulsar in the Background of Westerlund 1*, *Univ.*, 10, 274 (2024)
222. Ryu, T., **Perna, R.**, Cantiello, M., *Tidal Disruption Encores*, *Astrophys. J. Lett.*, 965, 25 (2024)
221. Tagawa, H., Kimura, S. S., Haiman, Z., **Perna, R.**, Bartos, I. *Shock cooling and break-out emission for optical flares associated with gravitational wave events*, *Astrophys. J.*, 966, 21 (2024)
220. Xin, C., Haiman, Z., **Perna, R.**, Wang, Y.-H., Ryu, T. *Tidal Peeling Events: low-eccentricity tidal disruption of a star by a stellar-mass black hole*, *Astrophys. J.*, 961, 149 (2024)
219. Ryu, T., de Mink, S., Farmer, R., Pakmor, R., **Perna, R.**, Springel, V. *Close Encounters of Star - Black Hole Binaries with Single Stars*, *MNRAS*, 527, 2734 (2024)
218. Ryu, T., Valli, R., Pakmor, R., **Perna, R.**, de Mink, S. E., Springel, V. *Close encounters of black hole - star binaries with stellar-mass black holes*, *MNRAS*, 525, 5752 (2023)
217. Soriano-Guerrero, C., Viganò, D., **Perna, R.**, Akgün, T., Palenzuela, C. *Magnetic winding and turbulence in ultra-hot Jupiters*, *MNRAS*, 525, 626 (2023)
216. Gottlieb, O., Issa, D., Jacquemin-Ide, J., Liska, M., Foucart, F., Tchekhovskoy, A., Metzger, B. D., Quataert, E., **Perna, R.**, Kasen, D., et al. *Large-scale Evolution of Seconds-long Relativistic Jets from Black Hole-Neutron Star Mergers*, *Astrophys. J. Lett.*, 954, 21 (2023)

215. Gottlieb, O., Issa, D., Jacquemin-Ide, J., Liska, M., Tchekhovskoy, A., Foucart, F., Kasen, D., **Perna, R.**, Quataert, E., Metzger, B. D. *Hourlong Near-UV/Optical Emission from Mildly Relativistic Outflows in Black Hole-Neutron Star Mergers*, *Astrophys. J. Lett.*, 953, 11 (2023)
214. Wang, Y., Ford, S., **Perna, R.**, McKernan, B., Zhu, Z., Zhang, B. *Effective two-body scatterings around a massive object*, *MNRAS*, 523, 2014 (2023)
213. Ye, C. S., Fragione, G., **Perna, R.** *On the Tidal Capture of White Dwarfs by Intermediate-mass Black Holes in Dense Stellar Environments*, *Astrophys. J.*, 953, 141 (2023)
212. Lazzati, D., **Perna, R.**, Gompertz, B., Levan, A. *GRB 191019A: a short gamma-ray burst in disguise from the disk of an active galactic nucleus*, *Astrophys. J. Lett.*, 950L, 20 (2023)
211. Dall’Osso, S., Stratta, G., **Perna, R.**, De Cesare, G., Stella, L. *Magnetar central engines in Gamma-Ray Bursts follow the universal relation of accreting magnetic stars*, *Astrophys. J. Lett.*, 949L, 32 (2023)
210. Tagawa, H., Kimura, S. S., Haiman, Z., **Perna, R.**, Bartos, I. *Observable signature of merging stellar-mass black holes in active galactic nuclei*, *Astrophys. J.*, 950, 13 (2023)
209. Ray, M., Lazzati, D., **Perna, R.** *The effects of Time-Variable Absorption due to Gamma-Ray Bursts In Active Galactic Nuclei Accretion Disks*, *MNRAS*, 521, 4233 (2023)
208. Tagawa, H., Kimura, S. S., Haiman, Z., **Perna, R.**, Bartos, I. *Observable signatures of stellar-mass black holes in active galactic nuclei*, *Astrophys. J. Lett.*, 946, 3 (2023)
207. Ryu, T., **Perna, R.**, Pakmor, R., Ma, J.-Z., Farmer, R. de Mink, S. E. *Close Encounters of Tight Binary Stars with Stellar-mass Black Holes*, *MNRAS*, 519, 5787 (2023)
206. Ibrahim, A. Y. et. al (includ. **Perna, R.**) *Deep X-ray and radio observations of the first outburst of the young magnetar Swift J1818.0-1607*. *Astrophys. J.*, 943, 20 (2023)
205. **Perna, R.**, Grishin, E. *Disks of Stars in the Galactic Center Triggered by Tidal Disruption Events*, *Astrophys. J. Lett.*, 939, 17 (2022) [Featured in AASNOVA Highlights and New Scientist Magazine]
204. Lazzati, D., Soares, G., **Perna, R.** *Prompt Emission of Gamma-Ray Bursts in the High-density Environment of Active Galactic Nuclei Accretion Disks*, *Astrophys. J. Lett.*, 938, 18 (2022)
203. Wang, Y.-H., Lazzati, D., **Perna, R.** *The emergence of diffused Gamma-Ray Burst afterglows from the disks of Active Galactic Nuclei*, *MNRAS*, 516, 5935 (2022)
202. Ryu, T., **Perna, R.**, Wang, Y.-H. *Close Encounters of Stars with Stellar-mass Black Hole Binaries*, *MNRAS*, 516, 2204 (2022)
201. Du, P., Egana-Ugrinovic, D., Essig, R., Fragione, G., **Perna, R.** *Searching for Ultra-light Bosons and Constraining Black Hole Spin Distributions with Stellar Tidal Disruption Events*, *Nature Comm.*, 13, 4626 (2022)

200. Yang, Y., Bartos, I., Fragione, G., Haiman, Z., Kowalski, M., Marka, S., **Perna, R.**, Tagawa, H. *Micro Tidal Disruption Events in Active Galactic Nuclei*, *Astrophys. J. Lett.*, 933, 28 (2022)
199. **Perna, R.**, Artale, M. C., Wang, Y.-H., Mapelli, M., Lazzati, D., Sgalletta, C., Santoliquido, F. *Host galaxies and electromagnetic counterparts to binary neutron star mergers across the cosmic time: Detectability of GW170817-like events*, *MNRAS*, 512, 2654 (2022)
198. Alford, J. A. J., Gotthelf, E. V., **Perna, R.**, Halpern, J. P. *Measuring the Non-Axially-Symmetric Surface Temperature Distribution of the Central Compact Object in Puppis A*, *Astrophys. J.*, 927, 233 (2022)
197. Tagawa, H., Kimura, S. S., Haiman, Z., **Perna, R.**, Tanaka, H., Bartos, I. *Can stellar-mass black hole growth disrupt disks of active galactic nuclei? The role of mechanical feedback*, *Astrophys. J.*, 927, 41 (2022)
196. Wang, Y.-H., **Perna, R.**, Leigh, N. W. C., Shara, M. M. *Hot Jupiter formation in dense clusters: secular chaos in multiplanetary systems*, *MNRAS*, 509, 5253 (2022)
195. Wang, Y.-H., McKernan, B., Ford, S., **Perna, R.**, Leigh, N. W. C., MacLow, M. M. *Symmetry Breaking in Dynamical Encounters in the Disks of Active Galactic Nuclei*, *Astrophys. J. Lett.*, 923, 23 (2021)
194. Lazzati, D., **Perna, R.**, Ciolfi, R., Giacomazzo, B., Lopez-Camara, D., Morsony, B. *Two steps forward and one step sideways: the propagation of relativistic jets in realistic binary neutron star merger ejecta*, *Astrophys. J. Lett.*, 918, 6 (2021)
193. **Perna, R.**, Tagawa, H., Haiman, Z., Bartos, I. *Accretion-Induced Collapse of Neutron Stars in the Disks of Active Galactic Nuclei*, *Astrophys. J.*, 915, 10 (2021)
192. Wang, Y.-H., Leigh, N., Liu, B., **Perna, R.** *SpaceHub: A high-performance gravity integration toolkit for few-body problems in astrophysics*, *MNRAS*, 505, 1053 (2021)
191. Borghese, A. et al. (includ. **Perna, R.**) *The X-ray evolution and geometry of the 2018 outburst of XTE J1810–197*, *MNRAS*, 504, 5244 (2021)
190. Jermyn, A. S., Dittmann, A. J., Cantiello, M., **Perna, R.** *Stellar Evolution in the Disks of Active Galactic Nuclei Produces Rapidly Rotating Massive Stars*, *Astrophys. J.*, 914, 105 (2021)
189. Wang, Y.-H., **Perna, R.**, Armitage, P. J. *Partial tidal disruption events by stellar mass black holes: Gravitational instability of stream and impact from remnant core*, *MNRAS*, 503, 6005 (2021)
188. Campana, S., Lazzati, D., **Perna, R.**, Bernardini, M. G., Nava, L. *The variable absorption in the X-ray spectrum of GRB 190114C*, *A&A*, 649, 135 (2021)
187. **Perna, R.**, Lazzati, D., Cantiello, M. *Electromagnetic Signatures of Relativistic Explosions in the Disks of Active Galactic Nuclei*, *Astrophys. J. Lett.*, 906, 7 (2021)
[Featured in AASNOVA Highlights]

186. Fragione, G., **Perna, R.**, Loeb, A. *Calibrating the binary black hole population in nuclear star clusters through tidal disruption events*, MNRAS, 500, 4307 (2021)
185. Wang, Y.-H., Leigh, N. W. C., **Perna, R.**, Shara, M. M. *Hot Jupiter and ultra-cold Saturn formation in dense star clusters*, Astrophys. J., 905, 536 (2020)
184. Dehman, C., Viganó, D., Rea, N., Pons, J. A., **Perna, R.**, Garcia-Garcia, A. *On the Rate of Crustal Failures in Young Magnetars*, Astrophys. J. Lett., 902, 32 (2020)
183. Leigh, N. W. C., Toonen, S., Portegies Zwart, S. F., **Perna, R.** *Mergers of Equal-Mass Binaries with Compact Object Companions from Mass Transfer in Triple Star Systems*, MNRAS, 496, 1819 (2020)
182. Lazzati, D., Ciolfi, R., **Perna, R.** *Intrinsic properties of the engine and jet that powered the short gamma-ray burst associated with GW170817*, Astrophys. J., 898, 59 (2020)
181. Wang, Y.-H., **Perna, R.**, Leigh, N. *Planetary Architectures in Interacting Stellar Environments*, MNRAS, 496, 1453 (2020)
180. Fragione, G. Metzger, B. D., **Perna, R.**, Leigh, N. W. C., Kocsis, B. *Electromagnetic transients and gravitational waves from white dwarf disruptions by stellar black holes in triple systems*, MNRAS, 495, 1061 (2020)
179. Esposito, P. et al. (includ. **Perna, R.**) *A very young radio-loud magnetar*, Astrophys. J. Lett. , 896, 30 (2020)
178. Rea, N. et al. (includ. **Perna, R.**) *The X-ray outburst of the Galactic Center magnetar over six years of Chandra observations*, Astrophys. J., 894, 159 (2020)
177. Wang, Y.-H., **Perna, R.**, Leigh, N. *Giant Planet Swaps during Close Stellar Encounters*, Astrophys. J. Lett., 891, 14 (2020)
176. Wang, Y.-H, Leigh, N., Sesana, A., **Perna, R.**, *The cosmological distribution of compact object mergers from dynamical interactions with SMBH binaries*, MNRAS, 490, 262 (2019)
175. Artkop, K., Smith, R., Corsi, A., Giacintucci, S., Peters, W. M., **Perna, R.**, Cenko, S. B., Clarke, T. E., *Radio follow-up of a candidate gamma-ray transient in the sky localization area of GW170608*, Astrophys. J., 489, 727 (2019)
174. Fragione, G., Leigh, N., **Perna, R.**, Kocsis, B. *Tidal disruption events onto stellar black holes in triples*, MNRAS, 489, 727 (2019)
173. Lazzati, D., **Perna, R.**, *Jet-cocoon outflows from neutron star mergers: structure, light curves, and fundamental physics*, Astrophys. J., 881, 89 (2019)
172. Fragione, G., Leigh, N. W. C., **Perna, R.** *Black hole and neutron star mergers in galactic nuclei: The role of triples*, MNRAS, 488, 2825 (2019)
171. Fragione, G., Grishin, E., Leigh, N. W. C., Perets, H. B., **Perna, R.** *Black hole and neutron star mergers in galactic nuclei*, MNRAS, 488, 47 (2019)

170. **Perna, R.**, Wang, Y.-H., Farr, W. M., Leigh, N., Cantiello, M., *Constraining the Black Hole Initial Mass Function with LIGO/VIRGO Observations*, *Astrophys. J. Letters*, 878, 1 (2019) [Featured in AASNOVA Highlights]
169. Scibelli, S., **Perna, R.**, Keeton, C. *Biases in inferring dark matter profiles from dynamical and lensing measurements*, *MNRAS*, 485, 5880 (2019)
168. **Perna, R.**, Lazzati, D., Farr, W. M., *Limits on electromagnetic counterparts of gravitational wave-detected binary black hole mergers*, *Astrophys. J.*, 875, 49 (2019)
167. Borghese, A. et al. (includ. **Perna, R.**), *The multi-outburst activity of the magnetar in Westerlund I*, *MNRAS*, 484, 2931 (2019)
166. Wang, Y.-H., Leigh, N., Sesana, A., **Perna, R.**, *Hypervelocity binaries from close encounters with a SMBH-IMBH binary: orbital properties and diagnostics*, *MNRAS*, 482, 3206 (2019)
165. Ryu, T., Zingale, M., **Perna, R.**, *Turbulence-driven thermal and kinetic energy flux in the atmospheres of hot Jupiters*, *MNRAS*, 481, 5517 (2018)
164. Lazzati, D., **Perna, R.**, Morsony, B. J., Lopez-Camara, D., Cantiello, M., Ciolfi, R., Giacomazzo, B., Workman, J. C., *Late time afterglow observations reveal a collimated relativistic jet in the ejecta of the binary neutron star merger GW170817*, *Phys. Rev. Lett.*, 120, 241103 (2018) [Chosen as Editors'Suggestion and featured in APS highlights]
163. Borghese, A., Coti Zelati, F., Esposito, P., Rea, N., De Luca, A., Bachetti, M., Israel, G. L., **Perna, R.**, Pons, J. A., *Gazing at the ultra-slow magnetar in RCW 103 with NuSTAR and Swift*, *MNRAS*, 478, 471 (2018)
162. **Perna, R.**, Chruslinska, M., Corsi, A., Belczynski, K., *Binary Black Hole Mergers within the LIGO Horizon: Statistical Properties and prospects for detecting Electromagnetic Counterparts*, *MNRAS*, 477, 4228 (2018)
161. Ibraginov, T., Leigh, N. W. C., Ryu, T., Panurach, T., **Perna, R.**, *When do star clusters become multiple star systems? II. Toward a half-life formalism with four bodies*, *MNRAS*, 477, 4213 (2018)
160. **Perna, R.**, Lazzati, D., Cantiello, M., *Ultra Long Gamma-Ray Bursts from the collapse of Blue Super Giant stars: an end-to-end simulation*, *Astrophys. J.*, 859, 48 (2018)
159. Esposito, P., Rea, N., Lazzati, D., Matsuura, M., **Perna, R.**, Pons, J. A. *Can a bright and energetic X-ray pulsar be hiding amid the debris of SN 1987A?*, *Astrophys. J.*, 857, 58 (2018)
158. Wang, Y.-H., Leigh, N., Yuan, Y.-F., **Perna, R.** *The fate of close encounters between binary stars and binary supermassive black holes*, *MNRAS*, 476, 4595 (2018)
157. Ryu, T., **Perna, R.**, Haiman, Z., Ostriker, J. P., Stone, N. C. *Interactions between multiple supermassive black holes in galactic nuclei: a solution to the final parsec problem*, *MNRAS*, 473, 3410 (2018)

156. Lazzati, D., Lopez-Camara, D., Cantiello, M., Morsony, B. J., **Perna, R.**, Workman, J. C. *Off-axis prompt X-ray transients from the cocoon of short gamma-ray bursts*, *Astrophys. J. Letters*, 848, 6 (2017)
155. Dall’Osso, S., **Perna, R.**, *The distribution of tilt angles in newly born NSs: role of interior viscosity and magnetic field*, *MNRAS*, 472, 2142 (2017)
154. Belczynski, K., Ryu, T., **Perna, R.**, Berti, E., Tanaka, T. L., Bulik, T. *On the likelihood of detecting gravitational waves from Population III compact object binaries*, *MNRAS*, 471, 4702 (2017)
153. Ryu, T., Leigh, N., W. C., **Perna, R.** *Formation of runaway stars in a star-cluster potential*, *MNRAS*, 470, 3049 (2017)
152. Ryu, T., Leigh, N., W. C., **Perna, R.** *An analytic method for identifying dynamically-formed runaway stars*, *MNRAS*, 470, 2 (2017)
151. Piro, A. L., Giacomazzo, B. **Perna, R.** *The Fate of Neutron Star Binary Mergers*, *Astrophys. J. Letters*, 844, 19 (2017)
150. Ryu, T., Leigh, N. W. C., **Perna, R.** *Numerical study of the $N = 4$ binary-binary scatterings in a background potential*, *MNRAS*, 467, 4447 (2017)
149. Ciolfi, R., Kastaun, W., Giacomazzo, B., Endrizzi, A., Siegel, D. M., **Perna, R.** *General relativistic magnetohydrodynamic simulations of binary neutron star mergers forming a long-lived neutron star*, *Phys. Rev. D*, 95, 3016 (2017)
148. Natale, G., Rea, N., Lazzati, D., **Perna, R.**, Torres, D. F., Girart, J. M. *Dust radiative transfer modelling of the infrared ring around the magnetar SGR 1900+14*, *Astrophys. J.*, 837, 9 (2017)
147. Dall’Osso, S., **Perna, R.**, Tanaka, T. L., Margutti, R. *Flares in Gamma-Ray Bursts: Disc Fragmentation and Evolution*, *MNRAS*, 464, 4399 (2017)
146. Margutti, R. et al. (includ. **Perna, R.**) *X-rays from the location of the Bactrian Transient ASASSN-15lh*, *Astrophys. J.*, 836, 25 (2017)
145. Kawamura, T., Giacomazzo, B., Kastaun, W., Ciolfi, R., Endrizzi, A., Baiotti, L., **Perna, R.** *Binary Neutron Star Mergers and Short Gamma-Ray Bursts: Effects of Magnetic Field Orientation, Equation of State, and Mass Ratio*, *Phys. Rev. D*, 94f4012 (2016)
144. Ryu, T., Tanaka, T. L., **Perna, R.**, Haiman, Z. *Intermediate-mass black holes from Population III remnants in the first galactic nuclei*, *MNRAS*, 460, 4122 (2016)
143. **Perna, R.**, Lazzati, D., Giacomazzo, B. *Short Gamma-Ray Bursts from the Merger of Two Black Holes*, *Astrophys. J. Lett.*, 821, 18 (2016)
142. Israel, G. L., et al. (includ. **Perna, R.**) *The discovery, monitoring and environment of SGR J1935+2154*, *MNRAS*, 457, 3448 (2016)

141. Dall’Osso, S., **Perna, R.**, Papitto, A., Bozzo, E., Stella, L. *The accretion regimes of a highly magnetised NS: the unique case of NuSTAR J095551+6940.8*, MNRAS, 457, 3076 (2016)
140. Rodriguez Castillo, G. A. et al. (includ. **Perna, R.**) *The Outburst Decay of the Low Magnetic Field Magnetar SWIFT J1822.3-1606: Phase-resolved Analysis and Evidence for a Variable Cyclotron Feature*, MNRAS, 456, 414 (2016)
139. Ryu, T., Tanaka, T. L., **Perna, R.** *Formation, disruption and energy output of Population III X-ray binaries*, MNRAS, 456, 223 (2016)
138. Tanaka, T. L., O’Leary, R. M., **Perna, R.** *The imprint of the cosmic supermassive black hole growth history on the 21 cm background radiation*, MNRAS, 455, 2619 (2016)
137. Rea, N., Gullon, M., Pons, J. A., **Perna, R.**, Dainotti, M. G., Miralles, J. A., Torres, D. F. *Constraining the GRB-magnetar model by means of the Galactic pulsar population*, Astrophys. J., 813, 92 (2015)
136. Gullón, M., Pons, J. A., Miralles, J. A., Viganó, D., Rea, N., **Perna, R.** *Population Synthesis of Isolated Neutron Stars with magneto-rotational evolution II: from radio-pulsars to magnetars*, MNRAS, 454, 615 (2015)
135. Guillot, S., **Perna, R.**, Rea, N., Viganó, D., Pons, J. A. *Modeling of the Surface Emission of the Low-Magnetic Field Magnetar SGR 0418+5729*, MNRAS, 452, 3357 (2015)
134. Greiner, J. et al. (includ. **Perna, R.**) *Gamma-Ray Bursts Trace UV Metrics of Star Formation over $3 < z < 5$* , Astrophys. J., 809, 76 (2015)
133. Giacomazzo, B., Zrake, J., Duffell, P., MacFadyen, A. I., **Perna, R.** *Producing Magnetar Magnetic Fields in the Merger of Binary Neutron Stars*, Astrophys. J., 809, 39 (2015)
132. Coti Zelati, F. et al. (includ. **Perna, R.**) *The X-ray outburst of the Galactic Centre magnetar SGR J1745-2900 during the first 1.5 year*, MNRAS, 449, 2685 (2015)
131. Dall’Osso, S., **Perna, R.**, Stella, L. *NuSTAR J095551+6940.8: a highly magnetised neutron star with super-Eddington mass accretion*, MNRAS, 449, 2144 (2015)
130. Trenti, M., **Perna, R.**, Jimenez, R. *The luminosity and stellar mass functions of GRB host galaxies: Insight into the metallicity bias*, Astrophys. J., 803, 103 (2015)
129. Dall’Osso, S., Giacomazzo, B., **Perna, R.**, Stella, L. *Gravitational Waves from Massive Magnetars formed in Binary Neutron Star Mergers*, Astrophys. J., 798, 25 (2015)
128. Kempton, E. M.-R., **Perna, R.**, Heng, K. *High Resolution Transmission Spectroscopy as a Diagnostic for Jovian Exoplanet Atmospheres: Constraints from Theoretical Models*, Astrophys. J., 795, 24 (2014)
127. Viganò, D., **Perna, R.**, Rea, N., Pons, J. A. *Spectral features in isolated neutron stars induced by inhomogeneous surface temperatures*, MNRAS, 443, 31 (2014)

126. **Perna, R.**, Duffell, P., Cantiello, M., MacFadyen, A. I. *The fate of fallback matter around newly born compact objects*, *Astrophys. J.*, 781, 119 (2014)
125. Lazzati, D., Villeneuve, M., Lopez-Camara, D., Morsony, B., **Perna, R.** *On the observed duration distribution of gamma-ray bursts from collapsars*, *MNRAS*, 436, 1867 (2013)
124. Rea, N. et al. (includ. **Perna, R.**) *A strongly magnetized pulsar within grasp of the Milky Way's supermassive black hole*, *Astrophys. J. Lett.*, 775, 34 (2013)
123. **Perna, R.**, Viganò, D., Pons, J. A., Rea, N. *The imprint of the crustal magnetic field on the thermal spectra and pulse profiles of isolated neutron stars*, *MNRAS*, 434, 2362 (2013)
122. Viganò, D., Rea, N., Pons, J. A., **Perna, R.**, Aguilera, D. N., Miralles, J. A. *Unifying the observational diversity of isolated neutron stars via magneto-thermal evolution models*, *MNRAS*, 434, 123 (2013)
121. Trenti, M., **Perna, R.**, Tacchella, S. *Gamma-Ray Burst and Star Formation Rates: The Physical Origin for the Redshift Evolution of Their Ratio*, *Astrophys. J. Lett.* 773, 22 (2013)
120. Giacomazzo, B., **Perna, R.** *Formation of Stable Magnetars from Binary Neutron Star Mergers*, *Astrophys. J. Lett.* 771, 26 (2013)
119. De Souza, R. S., Mesinger, A., Ferrara, A., Haiman, Z., **Perna, R.**, Yoshida, N. *Constraints on Warm Dark Matter models from high-redshift long gamma-ray bursts*, *MNRAS* (2013)
118. Rea, N., Israel, G. L., Pons, J. A., Turolla, R., Viganò, D., Zane, S., Esposito, P., **Perna, R.** et al, *The outburst decay of the low magnetic field magnetar SGR 0418+5729*, *Astrophys. J.*, 770, 65 (2013)
117. Esposito, P. et al. (includ. **Perna, R.**) *X-ray and radio observations of the magnetar Swift J1834.9-0846 and its dust-scattering halo*, *MNRAS*, 429, 3123 (2013)
116. van Adelsberg, M., **Perna, R.** *Time Dependent Radiative Transfer for Multi-Level Atoms using Accelerated Lambda Iteration* *MNRAS*, 429, 1407 (2013)
115. Giacomazzo, B., **Perna, R.**, Rezzolla, L., Troja, E., Lazzati, D. *Compact Binary Progenitors of Short Gamma-Ray Bursts*, *Astrophys. J. Lett.*, 762, 18 (2013)
114. Tanaka, T., **Perna, R.**, Haiman, Z. *X-ray emission from high-redshift miniquasars: self-regulating the population of massive black holes through global warming*, *MNRAS*, 425, 2974 (2012)
113. Giacomazzo, B., **Perna, R.** *General Relativistic Simulations of Accretion Induced Collapse of Neutron Stars to Black Holes*, *Astrophys. J. Lett.*, 785, 8 (2012)
112. Yan, T., **Perna, R.**, Soria, R. *The Effect of Fallback Disks on the Spectral and Timing Properties of Neutron Stars*, *MNRAS*, 423, 2451 (2012)

111. Rea, N. et al. (includ. **Perna, R.**) *A new low magnetic field magnetar: the 2011 outburst of Swift J1822.3-1606*, *Astrophys. J.*, 754, 27 (2012)
110. **Perna, R.**, Heng, K., Pont, F. *The Effects of Irradiation on Hot Jovian Atmospheres: Heat Redistribution and Energy Dissipation*, *Astrophys. J.*, 751, 59 (2012)
109. Trenti, M., **Perna, R.**, Levesque, E. M., Shull, J. M., Stocke, John T. *Gamma-Ray-Burst Host Galaxy Surveys at Redshift $z > 4$: Probes of Star Formation Rate and Cosmic Reionization*, *Astrophys. J. Lett.*, 749, 38 (2012)
108. **Perna, R.**, Ho, W. C. G., Verde, L., van Adelsberg, M., Jimenez, R. *Signatures of photon-axion conversion in the thermal spectra and polarization of neutron stars*, *Astrophys. J.*, 748, 116 (2012)
107. Bernardini, F., **Perna, R.**, Gotthelf, E. Israel, G. L. Rea, N., Stella, L. *Emission geometry, radiation pattern, and magnetic topology of the magnetar XTE J1810-197 in its quiescent state*, *MNRAS*, 418, 638 (2011)
106. Pons, J. A., **Perna, R.** *Magnetars vs. High Magnetic Field Pulsars: a Theoretical Interpretation of the Apparent Dichotomy*, *Astrophys. J.*, 741, 123 (2011)
105. **Perna, R.**, Pons, J. A. *A Unified Model of the Magnetar and Radio Pulsar Bursting Phenomenology*, *Astrophys. J. Lett.*, 727, 51 (2011)
104. Gotthelf, E. V., **Perna, R.**, Halpern, J. P. *Modeling the surface X-ray emission and viewing geometry of PSR J0821-4300 in Puppis A*, *Astrophys. J.*, 724, 1316 (2010)
103. **Perna, R.**, Menou, K., Rauscher, E. *Ohmic Dissipation in the Atmospheres of Hot Jupiters*, *Astrophys. J.*, 724, 313 (2010)
102. Israel, G. L., et al. (includ. **Perna, R.**) *The 2008 October Swift detection of X-ray bursts/outburst from the transient SGR-like AXP 1E 1547.0-5408*, *MNRAS*, 408, 1387 (2010)
101. **Perna, R.**, Menou, K., Rauscher, E. *Magnetic Drag on Hot Jupiter Atmospheric Winds*, *Astrophys. J.*, 719, 1421 (2010)
100. Posselt, B., Schreyer, K., **Perna, R.**, Sommer, M. W., Klein, B., Slane, P. *Submillimeter observations of RXJ1856.5-3754*, *MNRAS*, 405, 1840 (2010)
99. **Perna, R.** & MacFadyen, A. *Flare-less long Gamma-ray Bursts and the properties of their massive progenitor stars*, *Astrophys. J. Lett.* 710, 103 (2010)
98. Robinson, P. B., **Perna, R.**, Lazzati, D., van Marle, A. J. *The rest-frame ultraviolet spectra of GRBs from massive rapidly-rotating stellar progenitors*, *MNRAS*, 401, 88 (2010)
97. Antonelli, L. A., D'Avanzo, P., **Perna, R.**, et al. *GRB 090426: the farthest short gamma-ray burst?*, *Astron. & Astrophys. Lett.*, 507, 45 (2009)
96. van Adelsberg, M. & **Perna, R.** *Soft X-ray Polarization in Thermal Magnetar Emission*, *MNRAS*, 399, 1523 (2009)

95. **Perna, R.**, Keeton, C. R. *Gravitational Lensing of Anisotropic Sources*, MNRAS, 397, 1084 (2009)
94. D’Elia, V., Fiore, F., **Perna, R.**, et al. *UVES/VLT high resolution absorption spectroscopy of the GRB080330 afterglow: a study of the GRB host galaxy and intervening absorbers*, Astron. & Astrophys. 503, 437 (2009)
93. Rea, N., et al. (includ. **Perna, R.**), *The outburst evolution of the new magnetar candidate SGR 0501+4516*, MNRAS, 396, 2419 (2009)
92. D’Avanzo, P., et al. (includ. **Perna, R.**), *The optical afterglows and host galaxies of three short/hard gamma-ray bursts*, Astron. & Astrophys. 498, 711 (2009)
91. Bernardini, F., et al. (includ. **Perna, R.**) *From outburst to quiescence: the decay of the transient Anomalous X-ray Pulsar XTE J1810-197*, Astron. & Astrophys. 498, 195 (2009)
90. Mignani, R. P., et al. (includ. **Perna, R.**) *VLT/NACO near-infrared observations of the transient radio magnetar 1E 1547.0-5408*, Astron. & Astrophys., 497, 451 (2009)
89. D’Elia, V., Fiore, F., **Perna, R.**, et al. *The prompt, high resolution spectroscopic view of the "naked-eye" GRB080319B*, Astrophys. J., 694, 332 (2009)
88. Piranomonte, S., et al. (includ. **Perna, R.**) *Probing the complex environments of GRB host galaxies and intervening systems: high resolution spectroscopy of GRB050922C*, Astron. & Astrophys., 492, 775 (2008)
87. Piranomonte, S., et al. (includ. **Perna, R.**) *The short GRB070707 afterglow and its very faint host galaxy*, Astron. & Astrophys., 491, 183 (2008)
86. Rossi, E. M., **Perna, R.**, Daigne, F. *"Orphan" afterglows in the Universal Structured Jet Model for gamma-ray bursts*, MNRAS, 390, 675 (2008)
85. Soria, R., **Perna, R.** *The Oldest Supernovae: X-ray Emission from 1941C, 1959D, 1968D*, Astrophys. J., 683, 767 (2008)
84. Campana, S., et al. (includ. **Perna, R.**) *Outliers from the mainstream: how a massive star can produce a gamma-ray burst*, Astrophys. J. Lett., 638, 9 (2008)
83. Lazzati, D., **Perna, R.**, Begelman, M. C. *X-ray flares, neutrino cooled disks, and the dynamics of late accretion in GRB engines*, MNRAS, 388L, 15 (2008)
82. Heng, K., Lazzati, D., **Perna, R.**, Garnavich, P., Noriega-Crespo, A., Bersier, D., Matheson, T., Pahre, M. *A Direct Measurement of the Dust Extinction Curve in an Intermediate-Redshift Galaxy*, Astrophys. J., 681, 1116 (2008)
81. **Perna, R.**, Gotthelf, E. V. *Constraints on the Emission and Viewing Geometry of the Transient Anomalous X-Ray Pulsar XTE J1810-197*, Astrophys. J., 681, 522 (2008)
80. Mignani, R. P., Zaggia, S., De Luca, A., **Perna, R.**, Bassan, N., Caraveo, P. A. *Optical and Infrared Observations of the X-ray source 1WGA J1713.4-3949 in the G347.3-0.5 SNR* Astron. & Astrophys., 484, 457 (2008)

79. Testa, V., Rea, N., Mignani, R. P., Israel, G. L., **Perna, R.**, et al. *Near-infrared Observations of Magnetars: XTE J1810-197, 1RXS J1708-4009, 1E 1841-045 and SGR 1900+14* Astron. & Astrophys., 482, 607 (2008)
78. **Perna, R.**, Soria, R., Pooley, D., Stella, L. *How rapidly do neutron stars spin at birth? Constraints from archival X-ray observations of extragalactic supernovae*, MNRAS, 384, 1638 (2008)
77. Bozzo, E., Falanga, M., Papitto, A., Stella, L., **Perna, R.**, et al. *X-ray eclipse time delays in 4U 2129+47*, Astron. & Astrophys., 476, 301 (2007)
76. Rea, N., Nichelli, E., Israel, G. L., (includ. **Perna, R.**) *Very deep X-ray observations of the Anomalous X-ray Pulsar 4U 0142+614*, MNRAS, 381, 293 (2007)
75. Lo Curto, G., Mignani, R., **Perna, R.**, Israel, G. L. *Deep VLT infrared observations of X-ray Dim Isolated Neutron Stars*, Astron. & Astrophys., 473, 539 (2007)
74. Janiuk, A., Yuan, Y., R. **Perna, R.**, T. Di Matteo, T. *Instabilities in the Time-Dependent Neutrino Disc in Gamma-Ray Bursts*, Astrophys. J, 664, 1011 (2007)
73. Mignani, R., **Perna, R.**, Rea, N., Israel, G. L., Mereghetti, S., Lo Curto, G. *VLT/NACO observations of the High-Magnetic field radio pulsar PSR J1119-6127*, Astron. & Astrophys., 471, 265 (2007)
72. Israel, G., Campana, S., Dall’Osso, S., Munro, M. P., Cummings, J., **Perna, R.**, Stella, L. *The Post-Burst Awakening of the Anomalous X-Ray Pulsar in Westerlund 1*, Astrophys. J, 664, 457 (2007)
71. Heng, K., Lazzati, D., **Perna, R.** *Dust Echoes from the Ambient Medium of Gamma-Ray Bursts*, Astrophys. J, 662, 1119 (2007)
70. Kuranov, A. G., Popov, S. B., Postnov, K. A., Volonteri, M., **Perna, R.** *On the dynamical formation of accreting intermediate mass black holes*, MNRAS, 377, 835 (2007)
69. D’Elia, V. et al. (includ. **Perna, R.**) *UVES/VLT high resolution spectroscopy of GRB 050730 afterglow: probing the features of the GRB environment*, Astron. & Astrophys., 467, 629 (2007)
68. Falanga, M., Bozzo, E., Stella, L., Burderi, L., Di Salvo, T., **Perna, R.** *On the maximum efficiency of the propeller mass-ejection mechanism*, Astron. & Astrophys. Lett., 464, 807 (2007)
67. Lazzati, D., **Perna, R.** *X-ray flares and the duration of engine activity in gamma-ray bursts*, MNRAS, 375L, 46 (2007)
66. Campana, S., Lazzati, D., Ripamonti, E., **Perna, R.**, et al. *A Metal-rich Molecular Cloud Surrounds GRB 050904 at Redshift 6.3*, Astrophys. J. Lett., 654, 17 (2007)
65. Guetta, D., Fiore, F., D’Elia, V., **Perna, R.**, et al. *The variable X-ray light curve of GRB 050713A: the case of refreshed shocks*, Astron. & Astrophys., 461, 95 (2007)

64. Skinner, S. L., **Perna, R.**, Zhekov, S. *The Chandra X-ray spectrum of the 10.6 Pulsar in Westerlund 1: Testing the magnetar Hypothesis*, *astrophys. J.*, 653, 587 (2006)
63. Lazzati, D., **Perna, R.**, Flasher, J., Dwarkadas, V. V., Fiore, F. *Time-resolved spectroscopy of GRB 021004 reveals a clumpy extended wind*, *MNRAS*, 372, 1791 (2006)
62. Kupcu Yoldas A., Greiner, J., **Perna, R.** *Constraining the environment of GRB 990712 through emission line fluxes*, *Astron. & Astrophys.* 457, 115 (2006)
61. Belczynski, K., **Perna, R.**, Bulik, T., Kalogera, V., Ivanova, N., Lamb, D. Q. *A Study of Compact Object Mergers as Short Gamma-Ray Burst Progenitors*, *Astrophys. J.*, 648, 1110 (2006)
60. **Perna, R.**, Bozzo, E., Stella, L. *On the Spin-up/Spin-down Transitions in Accreting X-Ray Binaries*, *Astrophys. J.*, 639, 363 (2006)
59. Covino, S., et al. (includ. **Perna, R.**) *Optical emission from GRB 050709: a short/hard GRB in a star-forming galaxy*, *Astron. & Astrophys. Lett.*, 447, 5 (2006)
58. **Perna, R.**, Armitage, P. J., Zhang, B. *Flares in Long and Short Gamma-Ray Bursts: A Common Origin in a Hyperaccreting Accretion Disk*, *Astrophys. J. Lett.*, 636, 29 (2006)
57. Tagliaferri, G., et al. (includ. **Perna, R.**) *GRB 050904 at redshift 6.3: observations of the oldest cosmic explosion after the Big Bang*, *Astron. & Astrophys. Lett.*, 443, 1 (2005)
56. Stratta, G., **Perna, R.**, Lazzati, D., Fiore, F., Antonelli, L. A., Conciatore, M. L. *Extinction properties of the X-ray bright/optically faint afterglow of GRB 020405*, *Astron. & Astrophys.* 441, 83 (2005)
55. Granot, J., Ramirez-Ruiz, E., **Perna, R.** *Afterglow Observations Shed New Light on the Nature of X-Ray Flashes*, *Astrophys. J.*, 630, 1003 (2005)
54. Filliatre, P., et al. (includ. **Perna, R.**) *Out of the darkness: the infrared afterglow of the INTEGRAL burst GRB 040422 observed with the VLT*, *Astron. & Astrophys.*, 438, 793 (2005)
53. Israel, G., et al. (includ. **Perna, R.**) *Discovery and monitoring of the likely IR counterpart of SGR 1806-20 during the 2004 γ -ray burst-active state*, *Astron. & Astrophys. Lett.*, 438, 1 (2005)
52. Volonteri, M., **Perna, R.** *Dynamical evolution of intermediate mass black holes and their observable signatures in the nearby Universe*, *MNRAS*, 358, 913 (2005)
51. Fiore, F., D'Elia, V., Lazzati, D., **Perna, R.**, et al. *A flash in the dark: UVES/VLT high resolution spectroscopy of GRB afterglows*, *Astrophys. J.*, 624, 853 (2005)
50. Mesinger, A., **Perna, R.**, Haiman, Z. *Constraints on the small-scale power spectrum of density fluctuations from high-redshift gamma-ray bursts*, *Astrophys. J.*, 623, 1 (2005)
49. Janiuk, A. **Perna, R.**, Di Matteo, T., Czerny, B. *Evolution of a neutrino-cooled disc in gamma-ray bursts* *MNRAS*, 355, 950 (2004)

48. Pierpaoli, E., **Perna, R.** *Radio emission from early-type galaxies and cosmic microwave background experiments*, MNRAS, 354, 1005 (2004)
47. Guetta, D., **Perna, R.**, Stella, L., Vietri, M. *Are All Gamma-Ray Bursts Like GRB 980425, GRB 030329, and GRB 031203?*, Astrophys. J., Lett. 615, 73 (2004)
46. **Perna, R.**, Stella, L. *Young Crab-like Pulsars and Luminous X-Ray Sources in Starbursts and Optically Dull Galaxies*, Astrophys. J., 615, 222 (2004)
45. Frontera, F., Amati, L., Lazzati, D., Montanari, E., Orlandini, M., **Perna, R.**, et al. *A Decreasing Column Density during the Prompt Emission from GRB 000528 Observed with BeppoSAX*, Astrophys. J., 614, 310 (2004)
44. Rea, N., Testa, V., Israel, G. L., Mereghetti, S., **Perna, R.**, et al. *Correlated Infrared and X-ray variability of the transient Anomalous X-ray Pulsar XTE J1810-197*, Astron. & Astrophys., 425, 5, (2004)
43. **Perna, R.**, Gaensler, B. M. *Are Supershells Powered by Multiple Supernovae? Modeling the Radio Pulsar Population Produced by OB Associations*, Astrophys. J. 606, 326 (2004)
42. Israel, G. L., Rea, N., Mangano, V. Testa, V., **Perna, R.**, et al. *The identification of the likely IR counterpart to the anomalous X-ray pulsar XTEJ1810-197*, Astrophys. J. Lett, 603, 97 (2004)
41. Blackman, E., **Perna, R.** *Pulsars With Jets May Harbor Dynamically Important Disks*, Astrophys. J. Lett, 601, 71 (2004)
40. **Perna, R.**, McDowell, J., Menou, K., Raymond, J., Medvedev, M. *Chandra Observations of the Dwarf Nova WX Hyi in Quiescence*, Astrophys. J. 598, 545, 552 (2003)
39. **Perna, R.**, Narayan, R., Rybicki, G., Stella, L., Treves, A. *Bondi Accretion and the Problem of the Missing Isolated Neutron Stars*, Astrophys. J. 594, 936 (2003)
38. **Perna, R.**, Sari, R., Frail, D. *Jets in GRBs: Tests and Predictions for the Structured Jet Model*, Astrophys. J., 594, 379 (2003)
37. De Pasquale, M., Piro, L., **Perna, R.** et al. *A comparative study of the X-ray afterglow properties of optically bright and dark GRBs*, Astrophys. J. 592, 1018 (2003)
36. Israel, G. L., Covino, S., **Perna, R.**, et al. *The IR counterpart to the Anomalous X-Ray Pulsar 1RXS J1708.49-400910*, Astrophys. J. Lett. 589, 93 (2003)
35. Heyl, J., **Perna, R.** *Broadband modeling of GRB 021004*, Astrophys. J. Lett., 585, 13 (2003)
34. Lazzati, D., **Perna, R.** *Time-Dependent Photoionization in a Dusty Medium III: The effect of dust on the photoionization of metals*, MNRAS, 340, 649 (2003)
33. **Perna, R.**, Lazzati, D., Fiore, F. *Time-Dependent Photoionization in a Dusty Medium II: Evolution of Dust Distributions and Optical Opacities*, Astrophys. J, 585, 775 (2003)

32. **Perna, R.**, Lazzati, D. *Time-Dependent Photoionization in a Dusty Medium I: Code Description and General Results*, *Astrophys. J.*, 580, 261 (2002)
31. Di Matteo, T., **Perna, R.**, Narayan, R. *Neutrino Trapping and accretion models for Gamma-ray Bursts*, *Astrophys. J.*, 579, 706 (2002)
30. **Perna, R.**, Vietri, M. *A Self-similar Solution for the Propagation of a Relativistic Shock in an Exponential Atmosphere*, *Astrophys. J. Lett.*, 569, 47 (2002)
29. **Perna, R.**, Belczynski, K. *Short GRBs and Mergers of Compact Objects: Observational Properties*, *Astrophys. J.*, 570, 252 (2002)
28. Menou, K., **Perna, R.**, Hernquist, L. *Hydrogen-Poor Disks in Compact X-Ray Binaries*, *Astrophys. J. Lett.*, 564, 81 (2002)
27. Lazzati, D., **Perna, R.** *Determining the location of Gamma-Ray Bursts through the Evolution of their Soft X-ray Absorption*, *MNRAS*, 330, 383 (2002)
26. Di Matteo, T., **Perna, R.**, Abel, T., Rees, M. J. *Radio Foregrounds for the 21cm Tomography of the Neutral Intergalactic Medium at High Redshifts*, *Astrophys. J.*, 564, 576 (2002)
25. Menou, K. **Perna, R.**, Hernquist, L. *Stability and Evolution of Supernova Fallback Disks*, *Astrophys. J.*, 559, 1032 (2001)
24. **Perna, R.**, Heyl, J. S., Hernquist, L. E., Juett, A. M., Chakrabarty, D. *Anomalous X-Ray Pulsars and Soft Gamma-Ray Repeaters: Spectral Fits and the Magnetar Model*, *Astrophys. J.*, 557, 18 (2001)
23. Lazzati, D., **Perna, R.**, Ghisellini, G. *Time-dependent Photoionization Opacities in Dense Gamma-ray Burst Environments*, *MNRAS*, 325, L19 (2001)
22. Ferrara, A., **Perna, R.** *Scintillation as a Probe of the Intergalactic Medium*, *MNRAS*, 325, 164 (2001)
21. Menou, K., **Perna, R.**, Hernquist, L. *Disk-Assisted Spindown of Young Radio Pulsars*, *Astrophys. J. Lett.*, 554, 63 (2001)
20. **Perna, R.**, Heyl, J., Hernquist, L. *X-ray Emission from Middle-aged Pulsars*, *Astrophys. J.*, 553, 809 (2001)
19. Menou, K., **Perna, R.**, Raymond, J. *X-Ray Lines from Hot Flows around White Dwarfs. Application to SS Cygni*, *Astrophys. J.*, 549, 509 (2001)
18. **Perna, R.**, Hernquist, L. *Disks Irradiated by Beamed Radiation from Compact Objects*, *Astrophys. J. Lett.*, 544, 57 (2000)
17. **Perna, R.**, Aguirre, A. *Gamma-ray burst afterglows as probes of galactic and intergalactic dust*, *Astrophys. J.*, 546, 56 (2000)
16. **Perna, R.**, Di Matteo, T. *Synchrotron Emission from Hot Accretion Flows and the Cosmic Microwave Background Anisotropy*, *Astrophys. J.*, 542, 68 (2000)

15. **Perna, R.**, Raymond, J., Narayan, R. *X-Ray Line Diagnostics of Hot Accretion Flows around Black Holes*, *Astrophys. J.*, 541, 898 (2000)
14. **Perna, R.**, Hernquist, L., Narayan, R. *Emission Spectra of Fallback Disks around Young Neutron Stars*, *Astrophys. J.*, 541, 344, (2000)
13. **Perna, R.**, Raymond, J. *Testing the Multiple Supernovae versus Gamma-Ray Burst Scenarios for Giant H I Supershells*, *Astrophys. J.* 539, 706 (2000)
12. **Perna, R.**, Heyl, J., Hernquist, L. *Consequences of Interstellar Absorption for Models of Anomalous X-Ray Pulsars*, *Astrophys. J. Lett.*, 538, 159 (2000)
11. **Perna, R.**, Raymond, J., Loeb, A., *Identifying Gamma-Ray Burst Remnants in Nearby Galaxies*, *Astrophys. J.*, 533, 685 (2000)
10. **Perna, R.**, Loeb, A. *Constraining the Beaming of γ -ray Bursts With All-Sky Radio Surveys*, *Astrophys. J. Lett.*, 509, 85 (1998)
9. **Perna, R.**, Loeb, A. *X-Ray Absorption by the Hot Intergalactic Medium*, *Astrophys. J. Lett.*, 503, 135 (1998).
8. Loeb, A., **Perna, R.** *Are HI Supershells the Remnants of γ -ray Bursts?*, *Astrophys. J. Lett.*, 503, 35 (1998).
7. **Perna, R.**, Loeb, A. *Identifying the Environment and Redshift of γ -ray Burst Afterglows from the Time Dependence of Their Absorption Spectra*, *Astrophys. J.*, 501, 467 (1998).
6. Loeb, A., **Perna, R.** *Microlensing of γ -ray Burst Afterglows*, *Astrophys. J.*, 495, 597 (1998).
5. **Perna, R.**, Loeb, A. *Probing the Mass Fraction of MACHOs in Extragalactic Halos*, *Astrophys. J.*, 493, 523 (1998).
4. **Perna, R.**, Loeb, A. *Microlensing of Quasars by Stars in Their Damped Ly α Absorbers*, *Astrophys. J.*, 489, 489 (1997).
3. **Perna, R.**, Loeb, A., Bartelmann, M. *Effects of Dust on Gravitational Lensing by Spiral Galaxies*, *Astrophys. J.*, 488, 550 (1997).
2. Di Stefano, R., **Perna, R.** *Identifying Microlensing by Binaries*, *Astrophys. J.*, 488, 55 (1997).
1. De Filippo, S., Lubritto, C., **Perna, R.**, Siano, F. *An Alternative Field Theoretic Setting for Anyon Systems*, *J. Phys., A* 27, L305 (1994)

Invited Articles and Book Chapters

4. **Perna, R.**, *A Heavyweight Merger*, *Physics Viewpoint*, *Physics*, 13, 111
3. Lazzati, D. **Perna, R.** *Short Gamma-Ray Bursts: marking the Birth of Black Holes from Coalescing Compact Binaries*, in “Physics of Relativistic Compact Objects in Binaries: from Birth to Coalescence”, Ed. M. Colpi, P. Casella, V. Casini, U. Moschella, A. Possenti, Springer (2008)

2. **Perna, R.** *The interaction of GRBs with their Environment*, Mod. Phys. Lett. A, Vol. 18, No. 37, 2611 (2003)
1. Lazzati, D., **Perna, R.** *Time-Dependent Effects of a Bright Source on the Interstellar Medium*, in “Astrophysics Research”, Ed. L. V. Ross (2003)

Conference Review Article

1. Andersson, N., et al. (includ. **Perna, R.**) *The Transient Gravitational-Wave Sky*, Class. and Quant. Grav., 30, 2002 (2013)

Articles in Conference Proceedings

32. Ryu, T., Tanaka, T., **Perna, R.** *Population III X-Ray Binaries*, Frontier Research in Astrophysics II, held in Mondello (Palermo), Italy (2016)
31. Viganó, D., Pons, J. A. , **Perna, R.** *Central Compact Objects in Magnetic Lethargy*, Proceedings of the MG13 Meeting on General Relativity. Edited by ROSQUIST KJELL ET AL. Published by World Scientific Publishing Co. Pte. Ltd., pp. 2304-2306 (2015)
30. **Perna, R.**, Pons, J. A., Viganó, D., Rea, N. *The Many Lives of Magnetized Neutron Stars*, Astronomische Nachrichten, 335, Issue 6-7, p.715 (2014)
29. Haiman, Z., Tanaka, T., **Perna, R.** *Self-regulating the early growth of black holes through global warming* in “First Stars iV - From Hayashi to the future - AIP Conference Proceedings, Volume 1480, pp. 303-308 (2012)
28. Pons, J. A. **Perna, R.** *Is the apparent dichotomy between bursting activity of magnetars and radio pulsars real ?*, Journ. of Phys. vol.343, Issue 1, pp. 012007 (2012)
27. Janiuk, A. Yuan, Y.-F., **Perna, R.**, Di Matteo, T. *Instabilities in the Gamma Ray Burst central engine. What makes the jet variable?* , in ”Jets at all scales”, Buenos Aires 2009, IAU Proceedings, 275, 349 , eds. G. Romero, R. Sunyaev, T. Belloni (2011)
26. D’Avanzo, P., et al. (includ. **Perna, R.**) *The optical afterglows and host galaxies of three short/hard gamma-ray bursts*, in “Probing stellar populations out to the distant Universe: Cefalu’ 2008, Proceedings of the International Conference. AIP Conference Proceedings, Volume 1111, pp. 524-527 (2009)
25. D’Elia, V., Fiore, F., Nicastro, F., **Perna, R.**, Krongold, Y. *High Resolution Spectroscopy of Gamma-Ray Burst Afterglows*, in “Probing stellar populations out to the distant Universe: Cefalu’ 2008, Proceedings of the International Conference. AIP Conference Proceedings, Volume 1111, pp. 495-502 (2009)
24. Soria, R., **Perna, R.**, Pooley, D., Stella, L. *How rapidly do neutron stars spin at birth?* to appear in the proceedings of the 10th Asian-Pacific Regional IAU Meeting (APRIM 2008), Kunming, China, Aug. (2008)
23. Bozzo, E., Falanga, M., Papitto, A., Stella, L., **Perna, R.**, et al. *X-ray eclipse time delays in 4U 2129+47* in ”The X-ray Universe 2008” Symposium held in Granada, Spain, 27-30 May, 2008, Published online at http://xmm.esac.esa.int/external/xmm_science/workshops/2008symposium, p.27 (2008)

22. Rossi, E. M., **Perna, R.**, Daigne, F. *Predictions for afterglows detected in surveys in the Universal Structured Jet Model*, Gamma-Ray Bursts 2007: Proceedings of the Santa Fe Conference. AIP Conference Proceedings, Volume 1000, pp. 444-447 (2008)
21. Lazzati, D., Morsony, B. C., **Perna, R.**, Begelman, M. C. *X-ray flares and their relation to the inner engine activity*, Gamma-Ray Bursts 2007: Proceedings of the Santa Fe Conference. AIP Conference Proceedings, Volume 1000, pp. 439-443 (2008)
20. van Adelsberg, M., Lazzati, D., **Perna, R.** *Ultra-Relativistic Shockwaves in Arbitrary Continuous Media I*, Astrophysics of Compact Objects: International Conference on Astrophysics of Compact Objects. AIP Conference Proceedings, Volume 968, pp. 36-39 (2008)
19. Mignani, R. P., Bagnulo, S., de Luca, A., Israel, G. L., Lo Curto, G., Motch, C., **Perna, R.**, Rea, N., Turolla, R., Zane, S. *Studies of neutron stars at optical/IR wavelengths*, Astrophys. and Space Science, 308, 203 (2007)
18. Kuranov, A. G., Popov, S. B., Postnov, K. A., Volonteri, M., **Perna, R.** *On the dynamic formation of accreting intermediate-mass black holes*, Astronomical and Astrophysical Transactions, vol. 26, Issue 1, p.87-89 (2007)
17. **Perna, R.** *The interplay between Gamma-Ray Bursts and their environment* Adv. in Space Res., Vol. 40, Issue 8, p. 1218-1223 (2007)
16. Janiuk, A., Yuan, Y., **Perna, R.**, di Matteo, T. *Neutrino Cooled disk in GRB central engine*, in AIP Conf. Proceedings, 801, 119-122 (2005).
15. Stratta, G., **Perna, R.**, Lazzati, D., Fiore, F., Antonelli, A., Conciatore, M. L. *Dust extinction properties of a sample of bright X-rays afterglows*, Il Nuovo Cimento C, 28, 693 (2005)
14. Stella, L., **Perna, R.** *Young Rotation-Powered Pulsars as Ultraluminous X-ray Sources in Star-Forming Galaxies* in AIP Conf. Proceedings, 797, 434 (2005)
13. Janiuk, A., Czerny, B., **Perna, R.**, di Matteo, T. *Hyper-Eddington accretion in GRB* Il Nuovo Cimento C, 28, 419 (2005)
12. **Perna, R.** *Radiative transfer in a dusty medium* Proceedings of the MG10 Meeting, World Scientific Publishing, 1404 (2005)
11. Bhargavi, S. G., Rhoads, J., **Perna, R.**, Feldmeier, J. & Greiner, J. *Searching for GRB remnants in nearby galaxies*, in AIP Conf. Proceedings, 727, 388 (2003)
10. Israel, G., et al. (includ. **Perna, R.**) *Unveiling the multiwavelength phenomenology of Anomalous X-ray Pulsars*, to appear in "Young Stars and their Environment", IAU Symposium 218, ASP Conf. Proceedings, Eds. F. Camilo & B. M. Gaensler (2004)
9. **Perna, R.** *Observational properties of Short GRBs in the Merger scenario*, Proceedings of the 2nd Workshop "Gamma-Ray Bursts in the Afterglow Era", ASP Conf. Series, 312, 393 (2004)

8. Bhargavi, S. G., Cowsik, R. & **Perna, R.** *Observing GRB remnants*, Proceedings of the 2nd Workshop "Gamma-Ray Bursts in the Afterglow Era", ASP Conf. Series, 312, 283 (2004)
7. Bhargavi, S. G., Rhoads, J., **Perna, R.**, Feldmeier, J., Greiner, J. *Searching for GRB Remnants in Nearby Galaxies*, Gamma-Ray Bursts: 30 Years of Discovery: Gamma-Ray Burst Symposium. AIP Conference Proceedings, Vol. 727, p. 388 (2004)
6. **Perna, R.** & Belckinski, K. *Observational properties of GRBs from Mergers*, Proceedings of the Conference "Gamma-ray Burst and Afterglow Astronomy", AIP Conf. Proceedings, 662, 417 (2003)
5. Lazzati, D. **Perna, R.** & Ghisellini, G. *X-ray spectroscopy of gamma-ray bursts: the path to the progenitor* Proceedings of the Conference "Gamma-Ray Burst and Afterglow Astronomy 2001", AIP Conf. Proceedings, 662, 411 (2003)
4. **Perna, R.** *Gamma-ray Burst Remnants: How can we find them?*, Proceedings of the 2nd Workshop "Gamma-Ray Bursts in the Afterglow Era", Rome, Oct.17-20, (2000)
3. **Perna, R.**, Heyl, J. S., Hernquist, L. E., Juett, A. M., Chakrabarty, D. *The thermal component of AXP and SGR spectra: fits with atmosphere models*, "Memorie della Societa' Astronomica Italiana", 73, 528, Eds: M. Feroci & S. Mereghetti (2002)
2. **Perna, R.**, Loeb, A., Raymond, J. *Effects of Gamma-ray Bursts on Their Gaseous Environment*, Proceedings of the "19th Texas Symposium on Relativistic Astrophysics and Cosmology", held in Paris, France, Dec. 14-18, Eds.: J. Paul, T. Montmerle, and E. Aubourg (CEA Saclay). (1998)
1. De Filippo, S., Lubritto, C., **Perna, R.**, Siano, F. *Are Anyons Relevant to Condensed Matter Physics? Maybe*, Proceedings of the Conference "Superconductivity and strongly correlated electrons", Vietri (Italy) (1994)

Selected Abstracts and Communications

10. Kempton, E., **Perna, R.**, Heng, K., Rauscher, E. *The Power of High Resolution Exoplanet Transmission Spectroscopy – Constraining Winds, Circulation, Tidal Locking, and Clouds*, American Astronomical Society, AAS Meeting #223, #230.06
9. Mignani, R., Israel, G., Rea, N., **Perna, R.**, Lo Curto, G. *Optical/near-IR observations of Magnetars*, 37th COSPAR Scientific Assembly. Held 13-20 July 2008, in Montreal, Canada., p.2032 (2008)
8. Gotthelf, Eric V., Halpern, J. P., **Perna, R.** *The Anatomy Of A Magnetar: XMM Monitoring Of Transient AXP XTE J1810-197* American Astronomical Society, HEAD meeting #9, #7.64, Bulletin of the American Astronomical Society, Vol. 38, p.360 (2006)
7. Garnavich, P., Pahre, M., Noriega-Crespo, A., Stanek, K. Z., Holland, S. T., Bersier, D., Matheson, T., **Perna, R.**, Krisciunas, K. *GRB 050525a, Mid-Infrared Observations*, GCN 3532 (2005)

6. Rea, N., et al. (includ. **Perna, R.**) *Correlated X-ray and IR decaying flux from the Anomalous X-ray Pulsar XTE J1810-197* Astron. Tel. 1 (2004)
5. Petrovic, N., Di Stefano, R., **Perna, R.** *Binary Sources and Binary Lenses in Microlensing Surveys of MACHOs* American Astronomical Society Meeting 203, #120.02 (2003)
4. Fiore, F., et al. (includ. **Perna, R.**) *GRB020813, high-resolution optical spectroscopy*, GCN 1524 (2002)
3. Savaglio, S., et al. (includ. **Perna, R.**) *GRB021004, High-resolution optical spectroscopy*, GCN 1633 (2002)
2. **Perna, R.**, Raymond, J., Narayan, R. *Thermal X-ray line emission from Hot Accretion Flows around Black Holes* American Astronomical Society, HEAD meeting #32, #31.20 (2000)
1. Menou, K., **Perna, R.**, Raymond, J. C, *Advection-Dominated Accretion Flows in Quiescent Dwarf Novae* American Astronomical Society, HEAD meeting #32, #41.05 (2000)